

1st Draft

NON-MARKOVIAN STOCHASTIC PROCESSES FROM GENOMICS TO ASTROPHYSICS

Christopher C. Bernido and Reinabelle C. Reyes

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Preface

This book deals with a large class of non-Markovian stochastic processes that have found applications in real-world datasets from genomics to astrophysics. The use of white noise analysis generates this large collection of non-Markovian stochastic processes which includes fractional Brownian motion as a special case. The procedure presented is useful in uncovering memory properties of any large dataset. Crucial to this wide application is the evaluation of the mean squared deviation (MSD) for any fluctuating values of the observables obtained from physical systems. This empirical MSD is then matched by a theoretical MSD of a stochastic model which leads to a closed form for the probability density function. The wide variety of possible memory functions is discussed, but the procedure can easily be extended to other novel memory behaviors not mentioned in this book. From the theoretical MSD which matches that obtained from the dataset, one could also proceed to evaluate the first passage time density of the stochastic process with memory. This allows us to obtain information about the probable time the stochastic observable would hit a threshold value given a particular memory behavior.

Extension to non-Gaussian stochastic processes using the same general approach is straightforward where a Mittag-Leffler measure is used. This also yields the Gaussian white noise stochastic processes with memory as a special case. A similarly broad collection of memory functions can be utilized for the non-Gaussian stochastic processes.

To encourage readers to dive into large datasets and uncover possible memory properties they possess, Chapter 4 discusses various applications ranging from DNA sequences all the way to pulsars. These examples give a flavor and template of the stochastic framework we use. Details of how these physical systems were specifically treated may be found in the journals they were published in as shown in the bibliography. Python codes necessary to process large datasets are also provided to make this book more friendly to those coming from different disciplines.

This book is based on a one semester graduate-level course taught in the past decade in three universities. Lectures on this topic were given at the National Institute of Physics, University of the Philippines, at the Mindanao State University-Iligan Institute of Technology, and at the University of San Carlos in Cebu City. As it turned out, this novel memory-dependent stochastic processes provided a rich source of graduate thesis material and publications. We thank the department heads in the three universities for the invitation to lecture on this topic.

We are grateful to Ludwig Streit and Takeyuki Hida for introducing us to stochastic processes and white noise analysis. Discussions with friends and academic colleagues deepened our understanding of the subject. Among them, we wish to mention Jose Luis da Silva, Tobias Kuna, Martin Grothaus, and Wolfgang Bock. Deep appreciation is also given to former students and collaborators especially Renante Violanda, Rev Aure, Allan Elnar, Christianlly Cena, Wilson Barredo, Pamela Roque, Matthew Escobido, Jinky Bornaes, Reynan Toledo, Lou Calotes, Gibson Maglasang, Karl Casas, and Prince Nayga. The authors also wish to acknowledge James David Tuñacao especially for sections 3.3 and 3.4. We wish to dedicate this book to the late Maria Victoria Carpio-Bernido who was there from the start.

Contents

Preface

GAUSSIAN PROCESSES WITH MEMORY

1. Introduction

- 1.1 Brief Overview: What is Memory?
- 1.2 Stochastic Differential Equation
- 1.3 White Noise Analysis: Basic Notions
- 1.4 Python Code: Fractional Brownian Paths

2. Fluctuations with Memory

- 2.1 Mathematical Representations of Fluctuations
- 2.2 Probability Density Function with Memory
- 2.3 Mean Square Deviation
- 2.4 Python Codes:
 - 2.4.1 Plotting the MSD
 - 2.4.2 Varying MSD Parameters

3. Detrending Datasets

- 3.1 Best-Fit Curves
- 3.2 Moving Averages
- 3.3 Power Spectral Density Analysis
- 3.4 Cycle Averaging Analysis: Fingerprinting

4. Matching Theory with Empirical Data

- 4.1 Empirical Mean Square Deviation
- 4.2 Ergodic Systems
- 4.3 Matching Mean Square Deviations
 - 4.3.1 Modulating the Amplitude
 - 4.3.2 Rescaling Time
- 4.4 Empirical Probability Displacement Distribution
- 4.5 Theoretical and Empirical Probability Distributions
 - 4.5.1 Drift Term
 - 4.5.2 Non-Gaussian Behavior

5. First Passage Time Density

- 5.1 Dirichlet Boundary Condition
- 5.2 Survival Probability Density
- 5.3 First Passage Time
- 5.4 Python Codes:

6. Applications

- 6.1 Nucleotide Distribution in a Genome
- 6.2 Codon Distribution in a Genome
- 6.3 Diffusion of Proteins of Varying Lengths
- 6.4 Ageing of Fibrin
- 6.5 Earthquakes in the Pacific Ring of Fire
- 6.6 Great Barrier Reef Degradation
- 6.7 Atmospheric CO₂ and Sea Surface Temperatures
- 6.8 Global Net Ecosystem Exchange
- 6.9 Solar Cycles and Sunspot Numbers
- 6.10 Pulsars and Timing Residuals

NON-GAUSSIAN PROCESSES WITH MEMORY

7. Non-Gaussian Stochastic Model

- 7.1 Mittag-Leffler Function and M -Wright Function
- 7.2 Mittag-Leffler Measure
- 7.3 Non-Gaussian Probability Density Function
- 7.4 Mean Square Deviation: Gaussian and non-Gaussian
- 7.5 Example: Pulsar Timing Residuals at Higher Lag Times
- 7.6 Fluctuations with Circular Geometry

Bibliography

Index

Chapter 1

Introduction

This chapter deals with the basic concepts and mathematical tools needed to discuss stochastic processes with memory. In particular, we would introduce a wide class of stochastic processes with memory using a memory modulated white noise framework which has found applications from biophysical systems to astrophysical objects. Having in mind readers coming from an interdisciplinary background, we shall introduce only those mathematical concepts that are required to enable us to develop a logical framework for handling large datasets. Specifically, we would like to uncover if a dataset has a memory or not, and what type of memory function is appropriate. For those who are mathematically inclined, we give the references in the bibliography for a more in-depth reading.

1.1 Brief Overview: What is Memory?

Recorded observations of our environment are replete with uncertainties. These uncertainties often arise from the many intractable interactions that any single entity under observation experiences. Uncontrolled fluctuations are ubiquitous from our understanding of sub-atomic quarks to experimental observations of events at the galactic scales. Slight errors are seemingly inherent in measurements that we perform. Despite the complex interactions in our environment, however, our mathematical description of various phenomena captures an essential part of reality. This has given birth to disciplines such as statistical physics, probability theory, and the theory of stochastic processes. In this book, we focus on a family of stochastic processes which we apply to various datasets to uncover memory properties in the systems being studied.

For almost a century, Brownian motion was a centerpiece in treating and understanding random motions. Brownian motion originated from the work of Albert Einstein (1879-1955) in 1905 who was trying to explain the fluctuating motion of a pollen grain suspended in water first observed in 1828 by the botanist Robert Brown (1773-1858). Brownian motion, also referred to as the Wiener process, is Markovian which means that it has no memory of the past. Preceding Einstein, however, was a similar mathematical treatment by Louis Bachelier (1870-1946) in his doctoral thesis which analyzed in 1900 the random movement of stocks prices. Although appreciated only much later, this work of Bachelier is now acknowledged as the birth of modern mathematical finance.

Brownian motion was later generalized to fractional Brownian motion, (fBm) which possesses a memory parametrized by an exponent H called the Hurst exponent in honor of Harold E. Hurst (1880-1978). Memory means that a stochastic process remembers and is correlated positively or negatively with its past. Studying the water levels of the Nile River for sixty years for the Egyptian government, Hurst discovered that water levels were correlated with past observations (Hurst 1951, Sutcliffe et al 2016). The Hurst exponent ranges from 0 to 1, and yields ordinary Brownian motion when, $H = 1/2$ (see, Fig. 1.1). Subdiffusive paths have, $0 < H < 0.5$, where successive points are negatively correlated. Superdiffusion, on the other hand, occurs when, $0.5 < H < 1$, where consecutive points are positively correlated. The value $H = 0.5$ yields ordinary

Brownian motion where the diffusion is memoryless (Mishura 2008, Shevchenko 2015). Negative correlation means that if the path goes to the right, the system remembers and there is greater probability that it will go left the next time. Positively correlated paths mean that if it goes right, there is a higher probability that it goes right again the next time leading to superdiffusion.

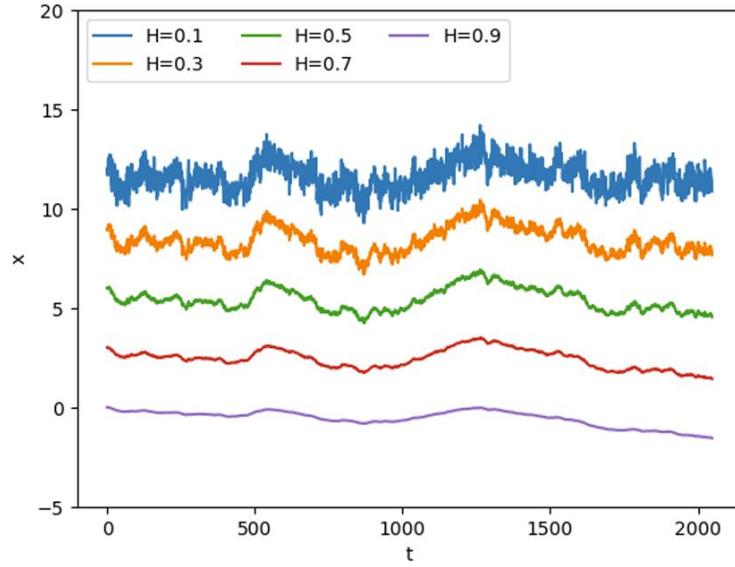


Figure 1.1. Paths of fractional Brownian motion for different values of the Hurst exponent H . Subdiffusive paths have, $0 < H < 0.5$. Superdiffusive paths have $0.5 < H < 1$. Only ordinary Brownian motion ($H = 0.5$), has no memory. Top to bottom: $H = 0.1, H = 0.3, H = 0.5, H = 0.7$, and $H = 0.9$.

Beyond fBm, there is a plethora of stochastic processes with memory which employ fractional calculus in their evaluation (Metzler et al 2014, Thapa et al 2018). Examples of these stochastic processes are given in Table 1.1. We shall focus on the last entry of the Table which has found wide-ranging applications in physical and social systems.

Table 1.1. Examples of different types of stochastic processes.

TYPE OF DIFFUSION	REFERENCES
Brownian Motion	Nelson 2001; Einstein 1905,
Scaled Brownian Motion	Jeon et al 2014
Fractional Brownian Motion	Hurst 1951; Mishura 2008,
Fractional Langevin motion	Deng and Barkai 2009
Continuous Time Random Walk	Scher and Montroll 1975; Metzler and Klafter 2000
Quenched Trap/Patch models	Massignan et al 2014; Miyaguchi et al 2011
Lévy Walk	Froemberg and Barkai 2013
Lévy Flight	Vahabai et al 2013
Fractal Environment	Meroz et al 2010; Weigel et al 2011
Memory Modulated White Noise	Bernido and Carpio-Bernido 2012, 2015

In our search for the truth in nature, we are like the famous story of three blind men who never encountered an elephant. Each blind man then touched a part of the elephant. One touched the side and said an elephant is like a wall. The other touched the elephant's trunk and said an elephant is like a tree. The last touched the tail and remarked that an elephant is like a rope. Many physical systems investigated have revealed different types of stochastic processes beyond ordinary Brownian motion (Metzler et al 2014, Thapa et al 2018). Depending on whether we deal with biological systems, astrophysical objects, the weather, or social dynamics, our perception of stochasticity in nature may vary (Fig. 1.2).

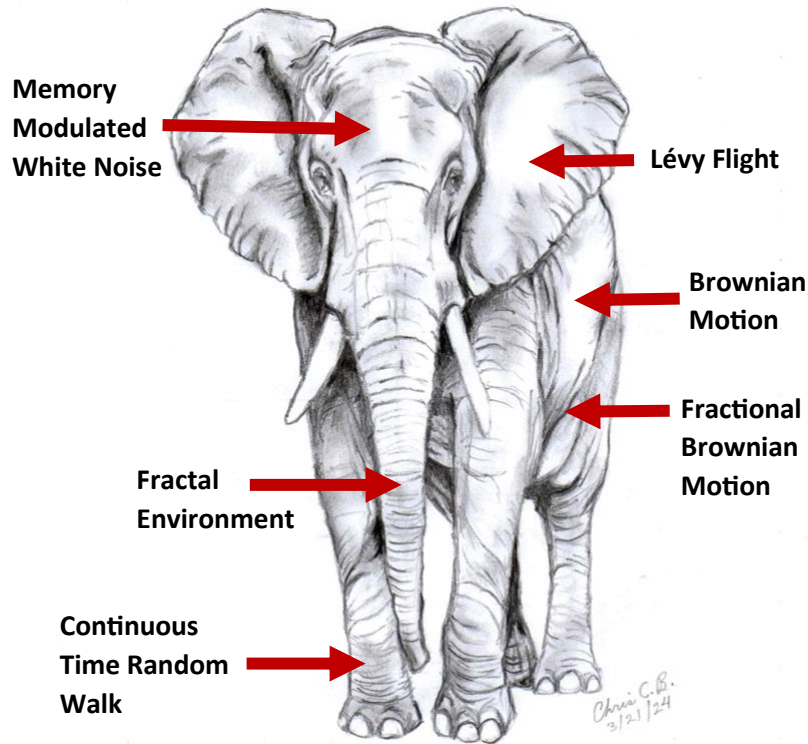


Figure 1.2 Depending on which part of reality we are probing, a different stochastic process may manifest.

We begin our discussion of stochastic processes with a stochastic differential equation followed by some mathematical preliminaries without going deep into the mathematical foundations which could be accessed elsewhere in the literature.

1.2 Stochastic Differential Equation

Most are familiar with the random walk where future directions are independent of the past. A continuous time version of the random walk is called the Brownian motion $B(\tau)$, also referred to as the Wiener process $W(\tau)$, where τ is time. Given a Brownian particle, its movement is influenced essentially by two factors: a deterministic component and random fluctuations affecting its position. Let us designate $dX(\tau)$ as the increment describing the particle's position, where τ is time, and write,

$$dX(\tau) = a_1(\tau)d\tau + a_2(\tau) dB(\tau) . \quad (1.1)$$

Equation (1.1) is a stochastic differential equation where the first term on the right-hand-side is deterministic while the second term is random with $B(\tau)$ a Brownian motion. The $a_1(\tau)$ is a time-dependent drift coefficient, and $[a_2(\tau)]^2$ is referred to as a diffusion coefficient. If we want to solve for $X(\tau)$, we integrate Eq. (1.1),

$$\int_0^t dX(\tau) = \int_0^t a_1(\tau)d\tau + \int_0^t a_2(\tau) dB(\tau) , \quad (1.2)$$

which leads to,

$$X(t) = X(0) + \int_0^t a_1(\tau)d\tau + \int_0^t a_2(\tau) dB(\tau) . \quad (1.3)$$

The last term, $\int_0^t a_2(\tau)dB(\tau)$, which is an integration over the Brownian motion has given rise to a long-standing controversy. After all, a different calculus is needed to handle Brownian motion which is continuous but nowhere differentiable.

A procedure to handle the integration over $dB(\tau)$ was first introduced in 1944 by Kiyosi Itô (1915-2008) which was mathematically consistent and lead to the result, $(dB)^2 = d\tau$, among others, which cannot be neglected. In some physical applications, the $(dB)^2 = d\tau$, has sometimes been referred to as a “spurious” term (McLaughlin and Schulman 1971, Brettschneider et al 2011). In 1966, Ruslan Stratonovich (1930-1997) gave a new representation for stochastic integrals and provided an equally mathematically consistent procedure for handling an integral over $dB(\tau)$, but gave a different answer from that of K. Itô. The Itô and Stratonovich procedures are different, but both are mathematically acceptable conventions even if they give different answers (van Kampen 1981, Mannella and McClintock 2012).

In the succeeding discussions, we bypass the Itô versus Stratonovich controversy by expressing the Brownian motion $B(\tau)$ in terms of the white noise random variable $\omega(\tau)$. White noise analysis is the framework of our non-Markovian stochastic treatment for many physical phenomena.

1.3 White Noise Analysis: Basic Notions

Our discussion of the basic concepts of white noise analysis starts with the Brownian motion $B(t)$ and some of its mathematical properties. Using the symbol $\langle \dots \rangle$ to represent the mean or expectation value, one obtains for $B(t)$,

$$\langle B(t) \rangle = 0 . \quad (1.4)$$

The correlation of the Brownian motion at two different times $B(t)$ and $B(s)$ can also be obtained. Multiplying $B(t)$ and $B(s)$, its mean is given by,

$$\langle B(t)B(s) \rangle = s , \quad (1.5)$$

for $s < t$.

Taking the derivative of $B(t)$ with respect to time, we can define the white noise random variable $\omega(t)$ as,

$$\omega(t) = dB(t)/dt . \quad (1.6)$$

The white noise variable $\omega(t)$ may be viewed as the velocity of the Brownian motion. It has a mean value of zero, i.e.,

$$\langle \omega(t) \rangle = 0 . \quad (1.7)$$

Two white noise variables at different times, however, are uncorrelated or independent from each other. This means that when multiplied together their mean is given by,

$$\langle \omega(t) \omega(s) \rangle = \delta(t - s) . \quad (1.8)$$

Equation (1.8) is always zero except when $t = s$.

In describing nature, we often encounter a model which is a functional of the Brownian motion $B(t)$ and its derivative $dB(t)$. To integrate terms containing $dB(t)$, one needs to specify whether the Itô formulation or the Stratonovich convention is being followed. Depending on the choice, the two results would be different, but both are mathematically acceptable as mentioned in the previous section. We can bypass this seeming dilemma in choice, by rewriting a model originally expressed in terms of the Brownian $B(t)$ and $dB(t)$, as a functional of the random white noise variable $\omega(t)$. Note that from Eq. (1.6) we can write,

$$dB(t) = \omega(t) dt \quad (1.9)$$

Integrating we get,

$$B(t) = \int \omega(t) dt \quad (1.10)$$

A physical phenomenon modelled or formulated in terms of the Brownian motion $B(t)$ and $dB(t)$ can, therefore, be reformulated in terms of the white noise variable $\omega(t)$ by using Eqs. (1.9) and (1.10). The motivation for re-expressing the equations in terms of the white noise variable $\omega(t)$ is that a mathematical framework already exists based on white noise analysis (Hida et al 1993, Kuo 1996, Obata 1994) which was founded in 1975 by Takeyuki Hida (1927-2017) who was a student of K. Itô.

White noise analysis is mathematically defined in infinite dimensional spaces (see, Table 1.2) using the embedding, $S \subset L^2 \subset S^*$, called the Gelfand triple. Here, the Hilbert space of square integrable functions L^2 links the space of test functions S with the space of Hida distributions S^* . Fundamental to the white noise framework is the Gaussian white noise measure $d\mu(\omega) \in S^*$, which allows us to handle and integrate functionals of the white noise variable $\omega(t)$. The $d\mu(\omega)$ is defined by the characteristic functional $C(\xi)$ which may be written as,

$$C(\xi) = \int_{S^*} e^{i \int_0^t \omega(\tau) \xi(\tau) d\tau} d\mu(\omega) = \exp \left(-\frac{1}{2} \int_0^t \xi^2 d\tau \right), \quad (1.11)$$

where ξ is a test function, $\xi \in S$.

An example of an integration involving the Gaussian white noise measure is the T -transform of a functional $\Phi(\omega)$ defined by,

$$T\Phi(\xi) \equiv \int_{S^*} e^{i \int_0^t \omega(\tau) \xi(\tau) d\tau} \Phi(\omega) d\mu(\omega) . \quad (1.12)$$

One could also introduce the S -transform which is related to the T -transform as,

$$S\Phi(\xi) = C(\xi) T\Phi(-i\xi) , \quad (1.13)$$

and,

$$T\Phi(\xi) = C(\xi) S\Phi(i\xi) , \quad (1.14)$$

where $C(\xi)$ is the characteristic functional, Eq. (1.11). For example, if one has a functional of the form,

$$\Phi(\omega) = \exp \left(i \int_0^t \omega(\tau) \eta(\tau) d\tau \right), \quad (1.15)$$

then its T -transform is given by, $C(\xi + \eta)$.

FINITE DIMENSIONS	INFINITE DIMENSION
Independent variable: x_j	 Independent random variable: $\omega(t)$
Coordinate system: $(x_1, \dots, x_n)$	 Coordinate system: $\{\omega(t); t \in \mathbb{R}\}$
Function: $f(x_1, \dots, x_n)$	 Functional: $\Phi(\omega(t); t \in \mathbb{R})$
Space: $\mathbb{R}^n$	 Space of Hida distributions: S^*
Lebesgue measure: dx	 Gaussian measure: $d\mu(\omega)$

Table 1.2. Correspondence between mathematical objects used in finite dimensions and their counterparts in infinite dimensions.

1.4 Python Code: Fractional Brownian Paths

Real-world applications of the memory modulated white noise framework often involve big datasets of sometimes several million data points. This is where a programming language such as Python comes in handy. Python codes will, therefore, be selectively included in our discussions to bridge the gap between theory and actual applications.

There may also be a need to simulate fluctuations to check outcomes. We start by numerically simulating the fractional Brownian motion as shown in Fig. 1.1. This is done using the Wood-Chan or circulant method discussed by Shevchenko (2015) for paths of the fractional Brownian motion. The Python code below is translated from the original Matlab code.

```
# import libraries
import numpy as np
import random
import matplotlib.pyplot as plt
%matplotlib inline

# define functions
def fbc(n,G):
    return ((n+1)**G + np.abs(n-1)**G - 2*n**G)/2.

def lambda_func(H,N):
    M = 2*N - 2
    C = np.zeros(M)
    G = 2*H
    for i in np.arange(N):
        # fill in first N out of M values of C (i=0 to N-1)
        C[i] = fbc(i,G)
    C[N:] = C[1:N-1][::-1]
    return np.real(np.fft.fft(C))**0.5

# set process parameters
q = 11
N = 2**q + 1
M = 2*N - 2
delta = 0.002
```

```

# initialize
noise = np.random.normal(size=(M))
fGnsamples = np.zeros((5,N))

# generate fractional Gaussian noise samples
for i in np.arange(5):
    H = 0.2*(i+1) - 0.1
    lambda_res = lambda_func(H,N)
    a = np.fft.ifft(noise)*lambda_res
    b = np.real(np.fft.fft(a))
    # apply offset in starting value for visualization
    fGnsamples[i,0] = 15-3*(i+1)
    fGnsamples[i,1:N] = delta**H*b[0:N-1]

# take cumulative sums to get the fractional Brownian motion samples
# cumulative sum over each row# cumulative sum over each row

import pickle
with open('fbm.pkl', 'wb') as file:
    pickle.dump(fBmsamples, file)

```

The two codes above are responsible for creating the “fbm.pkl” file which contains the dataset sample for fractional Brownian motion. Below is the code that plots the fractional Brownian paths, Fig.1.1.

```

# get values of H
Hval = (np.arange(5) + 1)*0.2 - 0.1

# plot fractional Brownian motion samples
for i in np.arange(len(Hval)):
    plt.plot(fBmsamples[:,i], label="H=%.1f" % Hval[i])
plt.ylim(-5, 20)
plt.legend(loc="upper left", ncol=3)
plt.xlabel("t")
plt.ylabel("x")

Output:

Text(0, 0.5, 'x')

```

Fluctuations with Memory

2.1 Mathematical Representations of Fluctuations

The complex interactions of many intractable factors in natural and social phenomena often lead to an observable that we can measure. Take for instance a country with several million citizens interacting which produces a number called the gross domestic product (GDP). Through the decades, a country's GDP would fluctuate up and down because of various economic policies, stability of the government, the public's mindset, world events, and countless other unpredictable factors which are impossible to tract individually. Likewise, the oscillations in stock prices are a byproduct of many interactions beyond anyone's control. The same holds true in the natural world. For instance, proteins migrating inside a cell are subjected to many intracellular forces and boundaries making its trajectory highly irregular. The many forces and influences affecting physical systems produce fluctuations resulting in uncertainties in our measurements. A series of recorded measurements, therefore, is often the combined result of both deterministic and stochastic components, the latter often referred to as noise.

To understand what is happening in our world, it is paramount that we could mathematically capture and represent the stochasticity around us. Let us consider Eq. (1.3) and write the derivative of the Brownian motion as, $dB(t) = \omega(t)dt$, where $\omega(t)$ is the random white noise variable. We have,

$$X(t) = X(0) + \int_{t_i}^t a_1(t')dt' + \int_{t_i}^t a_2(t')\omega(t') dt', \quad (2.1)$$

where the first integral on the right contains the drift term, and the second integral represents the random fluctuations where t_i and t are the fixed initial and final times, respectively. The $X(t)$ could represent any fluctuating dataset as exemplified by the physical systems discussed in Chapter 6. To make our mathematical representation more realistic, we can endow Eq. (2.1) with a memory by choosing $a_2(t')$ which modulates the random white noise $\omega(t)$ to have the form,

$$a_2(t, t') = g(t) f(t - t') h(t'), \quad (2.2)$$

where t is some final time. In Eq. (2.2), we refer to $f(t - t')$ as a memory function which depends on the elapsed time $(t - t')$ as t' varies from t_i to t in the integration. The $g(t)$ modulates the fluctuation and $h(t')$ is a window function. With Eq. (2.2), we can also write the fluctuating variable, Eq. (2.1), as

$$X(t) = X(0) + \int_{t_i}^t a_1(t')dt' + g(t) \int_{t_i}^t f(t - t') h(t') \omega(t') dt'. \quad (2.3)$$

We refer to this non-Markovian stochastic process as a memory modified white noise (MMWN) framework. The exact forms of the three functions $f(t - t')$, $g(t)$, and $h(t')$ will depend on the system under investigation (see, Chapter 6).

2.2 Probability Density Function with Memory

We now proceed to solve the probability density function (PDF) for a wide class of non-Markovian stochastic processes with varying memory properties by solving Eq. (2.3). Our approach uses the white noise calculus of T. Hida (1975, 1993) instead of the Itô stochastic calculus or Stratonovich stochastic approach.

In Eq. (2.3), the fluctuation has a fixed starting point at $X(0) = X_i$, but as time t' progresses, the modulated random white noise could end anywhere at final time t (Fig. 2.1). Suppose we are interested in a process which starts at X_i and ends at a chosen point, $X(t) = X_f$ at final time t , what would be its the probability density function? We can pin down the endpoint at a chosen point X_f (Fig. 2.2) using a Donsker delta function (Hida et al 1993, Kuo 1996) by writing,

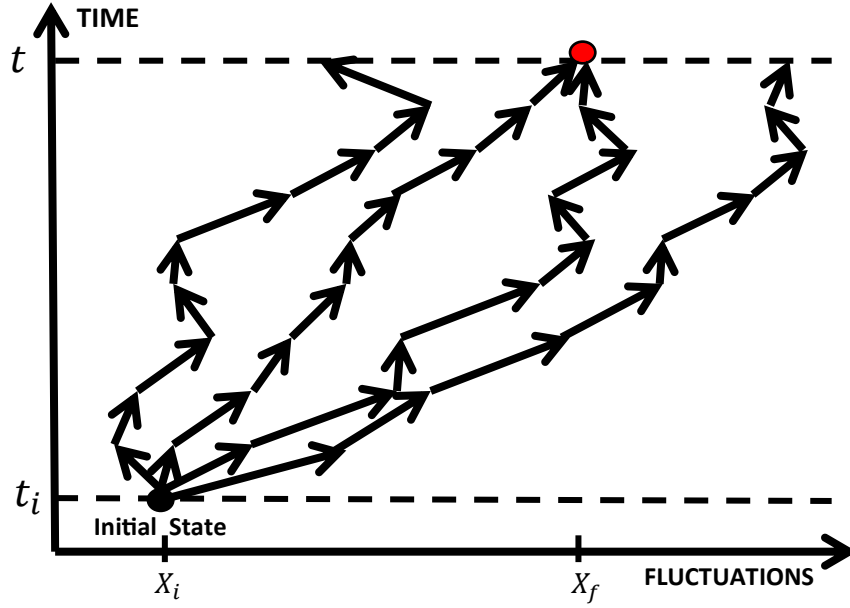


Figure 2.1. The fluctuation, Eq. (2.1), has a fixed initial point at $X(0) = X_i$ at initial time t_i , but it could end anywhere at final time t .

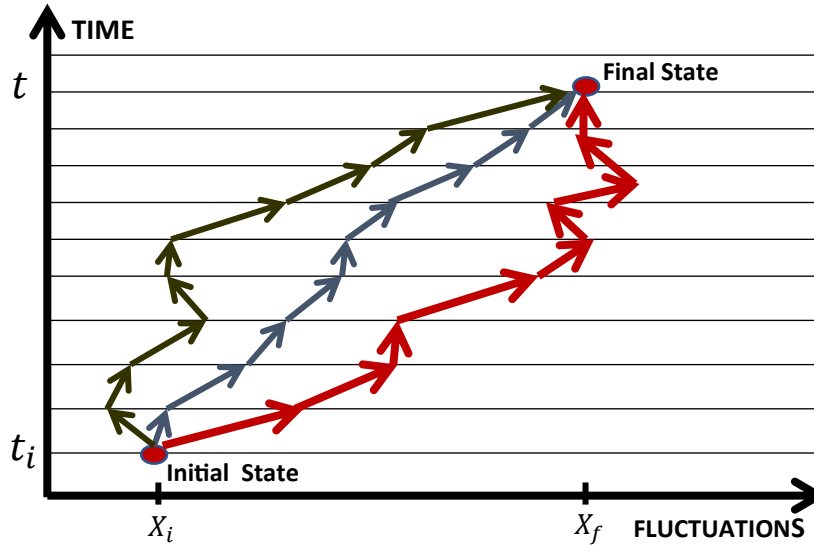


Figure 2.2. The endpoint of the fluctuation are pinned at a final value X_f using the Dirac delta function $\delta(X(t) - X_f)$. Since the delta function is always zero except at $X(t) = X_f$, the other paths not satisfying the delta function constraint have zero contribution.

$$\delta(X(t) - X_f) = \delta\left(X_i + \int_{t_i}^t a_1(t') dt' + \int_{t_i}^t a_2(t, t') \omega(t') dt' - X_f\right), \quad (2.4)$$

where we have used Eq. (2.1) to express $X(t)$. The delta function is always zero except for all paths which satisfy Eq. (2.4), i.e., those ending at the chosen endpoint X_f . We can then sum, or integrate, over all possible paths which begin at X_i and end at X_f to obtain the probability density function, $P(X_f, X_i; t)$, given by,

$$\begin{aligned} P(X_f, X_i; t) &= \int \delta(X(t) - X_f) d\mu(\omega) \\ &= \int \delta\left(X_i - X_f + \int_0^t a_1(t') dt' + \int_0^t a_2(t, t') \omega(t') dt'\right) d\mu(\omega). \end{aligned} \quad (2.5)$$

Here, we set, $t_i = 0$, and use the Gaussian white noise measure $d\mu(\omega)$ since the integrand in Eq. (2.5) is a white noise functional (Hida 1993, Kuo 1996). To facilitate the evaluation, we write the delta function in terms of its Fourier representation, i.e., $\delta(x) = (1/2\pi) \int_{-\infty}^{+\infty} e^{ikx} dk$, to get,

$$\begin{aligned} P(X_f, X_i; t) &= \frac{1}{2\pi} \int \int_{-\infty}^{+\infty} \exp\left[ik\left(X_i - X_f + \int_0^t a_1(t') dt' \right. \right. \\ &\quad \left. \left. + \int_0^t a_2(t, t') \omega(t') dt'\right)\right] dk d\mu(\omega). \end{aligned} \quad (2.6)$$

We need to perform the integrals over dk and $d\mu$ to obtain a closed form for the probability density function. To integrate over $d\mu(\omega)$, we rearrange the terms in Eq. (2.6) as,

$$\begin{aligned} P(X_f, X_i; t) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} dk \exp\left[ik\left(X_i - X_f + \int_0^t a_1(t') dt'\right)\right] \\ &\quad \times \int \exp\left[ik \int_0^t a_2(t, t') \omega(t') dt'\right] d\mu(\omega) \end{aligned} \quad (2.7)$$

We then let, $\xi(t') = ka_2(t, t')$, where, $\xi(t') \in S(\mathbb{R})$ for $S(\mathbb{R})$ the Schwartz space of test functions in the Gel'fand triple $S(\mathbb{R}) \subset L^2(\mathbb{R}) \subset S^*(\mathbb{R})$, with L^2 the Hilbert space of square integrable functions and $S^*(\mathbb{R})$ the space of tempered distribution (Hida et al 1993). The integration over $d\mu$ is facilitated by the characteristic functional $\mathcal{C}(\xi)$ (Hida et al 1993, Kuo 1996),

$$\mathcal{C}(\xi) = \int \exp\left(i \int_0^t \omega(t') \xi(t') d\tau\right) d\mu(\omega) = \exp\left(-\frac{1}{2} \int_0^t \xi^2(t') dt'\right). \quad (2.8)$$

With Eq. (2.8), we can carry out the integration over $d\mu(\omega)$ to get,

$$\begin{aligned} P(X_f, X_i; t) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} dk \exp\left[ik\left(X_i - X_f + \int_0^t a_1(t') dt'\right)\right] \\ &\quad \times \exp\left(-\frac{1}{2} k^2 \int_0^t a_2^2(t, t') dt'\right) dk. \end{aligned} \quad (2.9)$$

The integral over dk is a Gaussian integral which yields a closed form for the PDF,

$$P(X_f, X_i; t) = \frac{1}{\sqrt{2\pi \int_0^t a_2^2(t, t') dt'}} \exp\left(-\frac{\left(X_i - X_f + \int_0^t a_1(t') dt'\right)^2}{2 \int_0^t a_2^2(t, t') dt'}\right). \quad (2.10)$$

Using Eq. (2.2) which gives,

$$\int_0^t a_2^2(t, t') dt' = g(t)^2 \int_0^t [f(t - t') h(t')]^2 dt' , \quad (2.11)$$

we obtain an exact form for the PDF given by,

$$P(X_f, X_i; t) = \frac{1}{\sqrt{2\pi g(t)^2 \int_0^t [f(t - t') h(t')]^2 dt'}} \times \exp \left[\frac{-\left(X_i - X_f + \int_0^t a_1(t') dt'\right)^2}{2g(t)^2 \int_0^t [f(t - t') h(t')]^2 dt'} \right] \quad (2.12)$$

Only the memory function $f(t - t')$, modulating function $g(t)$, the window function $h(t')$, and the drift coefficient $a_1(t')$ remain to be specified which would depend on the system being investigated. Any square integrable product of the functions $f(t - t')$ and $h(t')$ solves the PDF exactly for a large class of non-Markovian stochastic processes (Bernido and Carpio-Bernido 2015) with a drift, $\int_0^t a_1(t') dt'$. Table 2.1 shows some of these functions, and physical systems with the time-dependent drift coefficient, $a_1(t') = 0$, are discussed in Chapter 6.

It is straightforward to check that Eq. (2.12) yields the PDF for ordinary Brownian motion. For zero drift, $a_1(t') = 0$, and $g(t) = h(t') = 1$, with, $f(t - t') = \sqrt{D}$, where D is the diffusion coefficient, Eq. (2.11) reduces to,

$$P(X_f, X_i; t) = \frac{1}{\sqrt{2\pi Dt}} \exp \left[-\frac{(X_i - X_f)^2}{2Dt} \right]. \quad (2.13)$$

Another example is fractional Brownian motion. For functions, $g(t) = h(t') = 1$, $f(t - t') = (t - t')^{H-(1/2)}/\Gamma(H + (1/2))$, and drift, $a_1(t') = 0$, Eq. (2.12) yields,

$$P(X_f, X_i; t) = \sqrt{\frac{H \Gamma^2(H + \frac{1}{2})}{\pi t^{2H}}} \exp \left[\frac{-H \Gamma^2(H + \frac{1}{2}) (x_1 - x_0)^2}{t^{2H}} \right], \quad (2.14)$$

which is the PDF of fractional Brownian motion in the Riemann-Liouville representation where, $0 < H < 1$, is the Hurst exponent (Calvo and Sanchez 2008). For $H = 1/2$, we obtain the ordinary Brownian motion, Eq. (2.13), with $D = 1$.

2.3 Mean Square Deviation

The theoretical mean square deviation can be evaluated using the PDF, Eq. (2.12) as, $M_X(t) = \langle X^2 \rangle - \langle X \rangle^2$, where $\langle \dots \rangle$ denotes the mean or ensemble average. For the fluctuating variable, we have ($n = 0, 1, 2, \dots$),

$$\langle X^n \rangle = \int_{-\infty}^{+\infty} X^n P(X, X_i) dX. \quad (2.15)$$

Given Eq. (2.12), we can write $\langle X \rangle$ explicitly as,

$$\langle X \rangle = \frac{\alpha}{\sqrt{\pi}} \int_{-\infty}^{+\infty} X e^{-\alpha^2 \left(X - X_i - \int_0^t a_1(t') dt' \right)^2} dX, \quad (2.16)$$

where we let,

$$\alpha^2 = \frac{1}{2g(t)^2 \int_0^t [f(t-t') h(t')]^2 dt'}. \quad (2.17)$$

To integrate Eq. (2.16) we let,

$$y = X - X_i - \int_0^t a_1(t') dt' \quad ; \quad dy = dX \quad (2.18)$$

and obtain,

$$\begin{aligned} \langle X \rangle &= \frac{\alpha}{\sqrt{\pi}} \int_{-\infty}^{+\infty} \left(y + X_i + \int_0^t a_1(t') dt' \right) e^{-\alpha^2 y^2} dy \\ &= \frac{\alpha}{\sqrt{\pi}} \int_{-\infty}^{+\infty} y e^{-\alpha^2 y^2} dy \\ &\quad + \frac{\alpha}{\sqrt{\pi}} \left(X_i + \int_0^t a_1(t') dt' \right) \int_{-\infty}^{+\infty} e^{-\alpha^2 y^2} dy \end{aligned} \quad (2.19)$$

The first integral is zero and we get,

$$\begin{aligned} \langle X \rangle &= 0 + \frac{\alpha}{\sqrt{\pi}} \left(X_i + \int_0^t a_1(t') dt' \right) \frac{\sqrt{\pi}}{\alpha} \\ &= X_i + \int_0^t a_1(t') dt'. \end{aligned} \quad (2.20)$$

Likewise, using Eqs. (2.17) and (2.18), we evaluate,

$$\langle X^2 \rangle = \frac{\alpha}{\sqrt{\pi}} \int_{-\infty}^{+\infty} X^2 e^{-\alpha^2 y^2} dy \quad (2.21)$$

From Eqs. (2.18) and (2.21), we have, $y = X - \langle X \rangle$, or $X^2 = y^2 + 2y\langle X \rangle + \langle X \rangle^2$. We could then write Eq. (2.21) as,

$$\begin{aligned} \langle X^2 \rangle &= \frac{\alpha}{\sqrt{\pi}} \int_{-\infty}^{+\infty} (y^2 + 2y\langle X \rangle + \langle X \rangle^2) e^{-\alpha^2 y^2} dy \\ &= \frac{\alpha}{\sqrt{\pi}} \int_{-\infty}^{+\infty} y^2 e^{-\alpha^2 y^2} dy \\ &\quad + \frac{\alpha 2\langle X \rangle}{\sqrt{\pi}} \int_{-\infty}^{+\infty} y e^{-\alpha^2 y^2} dy \\ &\quad + \frac{\alpha \langle X \rangle^2}{\sqrt{\pi}} \int_{-\infty}^{+\infty} e^{-\alpha^2 y^2} dy, \end{aligned} \quad (2.22)$$

where $\langle X \rangle$ is Eq. (2.20). With the second integral being zero, we are left with the first and last integral which give,

$$\begin{aligned}
\langle X^2 \rangle &= \frac{\alpha}{\sqrt{\pi}} \frac{\sqrt{\pi}}{2\alpha^3} + \frac{\alpha \langle X \rangle^2 \sqrt{\pi}}{\sqrt{\pi} \alpha} \\
&= \frac{1}{2\alpha^2} + \langle X \rangle^2 .
\end{aligned} \tag{2.23}$$

Equations (2.20) and (2.23) give us the mean square deviation,

$$\begin{aligned}
M_X(t) &= \langle X^2 \rangle - \langle X \rangle^2 \\
&= \frac{1}{2\alpha^2} + \langle X \rangle^2 - \langle X \rangle^2 = \frac{1}{2\alpha^2} ,
\end{aligned} \tag{2.24}$$

where the drift term understandably vanishes since it has nothing to do with the random deviations. With Eq. (2.17), the mean square deviation (MSD) contains the memory function, i.e.,

$$M_X(t) = g(t)^2 \int_0^t [f(t-t') h(t')]^2 dt' . \tag{2.25}$$

In many applications, the MSD is also referred to as second order structure function. Using Eq. (2.25), the probability density function, Eq. (2.12), can be written as,

$$P(X_f, X_i; t) = \frac{1}{\sqrt{2\pi M_X(t)}} \exp \left[-\frac{\left(X_i - X_f + \int_0^t a_1(t') dt' \right)^2}{2 M_X(t)} \right] \tag{2.26}$$

There is a plethora of possible forms of $M_X(t)$ where Eq. (2.25) is integrable which yields a wide class of probability density functions, Eq. (2.26), with a variety of memory behavior. Table 2.1 shows some of the different forms of memory function $f(t-t')$ and function $h(t')$ that yield the MSD, Eq. (2.25), for, $g(t) = 1$. We refer to this collection of non-Markovian stochastic processes as memory modified white noise (MMWN) which is the offshoot of modulating the white noise variable $\omega(t)$ as seen in the parametrization of the fluctuation, Eq. (2.3). To illustrate the effect of memory, Figure 2.3 shows sample plots of an MSD comparing the memoryless Brownian motion, the fractional Brownian motion for $H = 0.8$, and a MMWN given by entry number (12) of Table 2.1.

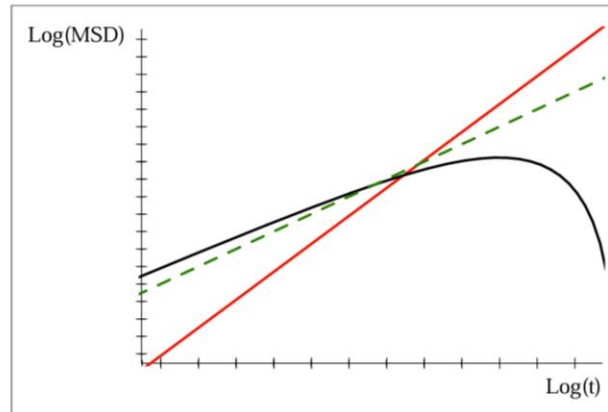


Figure 2.3. MSD plots comparing memoryless Brownian motion (green dashed-line); fractional Brownian motion with $H = 0.8$ (red solid line); and memory modulated white noise (MMWN) for $M_X(t)$ given by entry 12 of Table 2.1, with $\mu = 0.95$ and $a = 0.6$.

Table 2.1. Memory function with the corresponding Mean Square Deviation. The equation numbers are those found in Gradshteyn and Ryzhik (1994).

Memory Function $f(t - t')$	$h(t')$	MSD $M_X(t) = \langle X^2 \rangle - \langle X \rangle^2$
(1) $f = \frac{(t-t')^{H-1/2}}{\Gamma(H+\frac{1}{2})}$	$h = 1$	$\frac{t^{2H}}{2H \Gamma^2(H + \frac{1}{2})}$
(2) $f = \sqrt{bc} \exp\left[-\frac{b}{2}(t - t')\right]$	$h = 1$	$c[1 - \exp(-bt)]$
(3) $f = \sin^{\frac{1}{2}}(t - t')$	$h = \sqrt{J_0(t')}$	$t J_1(t)$ Eq. (6.674.7)
(4) $f = \cos^{\frac{1}{2}}(t - t')$	$h = \sqrt{J_0(t')}$	$t J_0(t)$ Eq. (6.674.8)
(5) $f = (t - t')^{\frac{\mu-1}{2}}$ [Re $\mu > 0$; $t > 0$]	$h = \frac{e^{-\beta/2t'}}{t'^{(\mu+1)/2}}$	$\Gamma(\mu)t^{\mu-1}\beta^{-\mu}e^{-\beta/t}$ Eq. (3.471.3)
(6) $f = (t - t')^{\frac{\mu-1}{2}}$ [Re $\mu > 0$; $t > 0$]	$h = \frac{e^{-\beta/2t'}}{t'^{(1-\nu)/2}}$ [Re $\beta > 0$]	$\frac{\Gamma(\mu)W_{\frac{1-2\mu-\nu}{2}, \frac{\nu}{2}}(\beta/t)}{\beta^{\frac{1-\nu}{2}} t^{\frac{1-2\mu-\nu}{2}} e^{\frac{\beta}{2t}}}$ Eq. (3.471.2)
(7) $f = (t - t')^{\frac{\mu-1}{2}}$ [Re $\mu > 0$; $t > 0$]	$h = \frac{e^{-\beta/2t'}}{t'^{\mu}}$ [Re $\beta > 0$]	$\frac{\Gamma(\mu)e^{-\frac{\beta}{2t}}K_{\mu-\frac{1}{2}}(\beta/2t)}{\sqrt{\pi t} \beta^{\mu-\frac{1}{2}}}$ Eq. (3.471.4)
(8) $f = (t - t')^{\frac{\mu-1}{2}}$ [Re $\mu > 0$]	$h = \frac{e^{\beta t'/2}}{t'^{(1-\nu)/2}}$ [Re $\nu > 0$]	$\frac{2B(\mu, \nu)_1 F_1(\nu; \mu + \nu; \beta t)}{t^{1-t-\nu}}$ Eq. (3.383.1)
(9) $f = (t - t')^{\frac{\mu-1}{2}}$ [Re $\mu > 0$]	$h = \frac{e^{\beta t'/2}}{t'^{(1-\mu)/2}}$	$\frac{\sqrt{\pi}\Gamma(\mu)I_{\mu-\frac{1}{2}}(\beta t/2)}{e^{-\beta t/2}(t/\beta)^{\frac{1}{2}-\mu}}$ Eq. (3.383.2)
(10) $f = (t - t')^{\frac{\mu-1}{2}}$ [Re $\mu > 0$]	$h = \frac{e^{-\beta t'/2}}{t'^{(1-\mu)/2}}$	$\frac{\sqrt{\pi}\Gamma(\mu)I_{\mu-\frac{1}{2}}(\beta t/2)}{e^{\beta t/2}(t/\beta)^{\frac{1}{2}-\mu}}$ Eq. (3.388.1)
(11) $f = (t - t')^{\frac{\mu-1}{2}}$ [Re $\mu > 0$]	$h = \frac{\sin^{\frac{1}{2}}(at)}{t'^{(1-\mu)/2}}$	$\frac{\Gamma(\mu)\sin(at/2)J_{\mu-\frac{1}{2}}(at/2)}{\pi^{-\frac{1}{2}} t^{\frac{1}{2}-\mu} a^{\mu-\frac{1}{2}}}$ Eq. (3.768.7)
(12) $f = (t - t')^{\frac{\mu-1}{2}}$ [Re $\mu > 0$]	$h = \frac{\cos^{\frac{1}{2}}(at)}{t'^{(1-\mu)/2}}$	$\frac{\Gamma(\mu)\cos(at/2)J_{\mu-\frac{1}{2}}(at/2)}{\pi^{-\frac{1}{2}} t^{\frac{1}{2}-\mu} a^{\mu-\frac{1}{2}}}$ Eq. (3.768.9)

(13) $f = (t - t')^{\frac{\mu-1}{2}}$ $[\text{Re } \mu > 0]$	$h = \frac{(t^2 + \beta^2)^{\frac{\nu}{2}}}{t^{\frac{1-\lambda}{2}}}$ $[\lambda > 0]$	$\frac{[\text{B}(\lambda, \mu)]^{-1} t^{1-\lambda-\mu}}{\beta^{2\nu} {}_3F_2\left(-\nu, \frac{\lambda}{2}, \frac{\lambda+1}{2}; \frac{\lambda+\mu}{2}, \frac{\lambda+\mu+1}{2}; -\frac{t^2}{\beta^2}\right)}$ $\left[\text{Re } \frac{t}{\beta} > 0\right]$; Eq. (3.254.1)
(14) $f = (t - t')^{-\frac{\nu}{2}}$ $[\text{Re } \nu < 1]$	$h = \sqrt{t'^\nu / (t' - c)}$ $[c < t']$	$\frac{\pi \left[1 - \cos(\nu\pi) \left(\frac{c}{t-c}\right)^\nu\right]}{\csc^{-1}(\nu\pi)}$ Eq. (3.228.1)
(15) $f = (t - t')^{-\frac{\nu}{2}}$ $[0.5 < \text{Re } \nu < 1]$	$h = \sqrt{\frac{t'^{\nu-1}}{(t' - c)}}$ $[c < t']$	$\frac{-\pi c^{\nu-1} \cot(\nu\pi)}{(t - c)^\nu}$ Eq. (3.228.2)
(16) $f = (t - t')^{\frac{\nu}{2}}$ $[\text{Re } \nu > -1 ; t > 0]$	$h = e^{-\mu t'/2}$	$\frac{(-\mu)^{-\nu-1} \Gamma(\nu + 1, -t\mu)}{e^{t\mu}}$ Eq. (3.382.1)
(17) $f = \sqrt{J_{1-\nu}(t - t')}$ $[-1 < \text{Re } \nu < 2]$	$h = \sqrt{J_\nu(t')}$	$J_0(t) - \cos(t)$ Eq. (11.3.38)
(18) $f = \sqrt{J_\nu(t - t')}$ $[\text{Re } \nu > -1]$	$h = \sqrt{t'^{-1} J_\mu(t')}$ $[\text{Re } \mu > 0]$	$J_{\mu+\nu}(t)/\mu$ Eq. (11.3.40)

It is straightforward to show that the PDF, Eq. (2.26), is a solution of the modified Fokker-Planck equation,

$$\frac{\partial P(X, X_i; t)}{\partial t} = -\mu(t) \frac{\partial P(X, X_i; t)}{\partial X} + \frac{1}{2} \left(\frac{\partial M_X(t)}{\partial t} \right) \frac{\partial^2 P(X, X_i; t)}{\partial X^2}, \quad (2.27)$$

where, $\mu(t) = \frac{\partial}{\partial t} \int_0^t a_1(t') dt'$. For example, using the $M_X(t)$ for fractional Brownian motion given by entry number (1) of Table 2.1, for zero drift, $a_1(t') = 0$, we get from Eq. (2.27) the result,

$$\frac{\partial P(X, X_i; t)}{\partial t} = \frac{t^{2H-1}}{2 \Gamma^2\left(H + \frac{1}{2}\right)} \frac{\partial^2 P(X, X_i; t)}{\partial X^2}. \quad (2.28)$$

Equation (2.28) is the modified diffusion equation for fractional Brownian motion in the Riemann-Liouville representation (Calvo and Sánchez 2008).

2.4 Python Codes

2.4.1 Plotting the MSD

To familiarize ourselves with the different mean square deviations, consider as an example, entry number (5) of Table 2.1. Choosing the coefficient in Eq. (2.25) as, $g(t)^2 = a$, we have,

$$M_X(t) = a \Gamma(\mu) t^{\mu-1} \beta^{-\mu} e^{-\beta/t}. \quad (2.29)$$

This MSD was used, for instance, to fit the empirical MSD of fluctuating sunspot numbers. Matching Eq. (2.29) with the empirical MSD for solar cycle 24,

the best-fit parameter values would be, $a = 3708.31$ (error 1230.48), $\mu = 1.02$ (error: 0.03), and $\beta = 2.43$ (error: 0.56). The plot of the MSD, Eq. (2.29), is shown in Fig. 2.4 for the following Python code.

Plotting an MSD

```
In: # import libraries
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
from scipy.special import gamma

# define MSD function
def msd(t, a, beta, mu):
    return a*gamma(mu)*t**(mu-1)*beta**(-1*mu)*np.exp(-1*beta/t)

a, err_a = 3708.31, 1230.48
beta, err_beta = 2.43, 0.56
mu, err_mu = 1.02, 0.03

# plot MSD (best-fit)

t = np.logspace(0, 3, 100) # equally log-spaced grid in time
m0 = msd(t, a, beta, mu)

plt.plot(t, m0, color='k')
plt.xscale("log")
plt.yscale("log")
plt.xlabel("t (days)")
plt.ylabel("MSD")
plt.title("Best-fit MSD")

Out: Text(0.5,1,'Best-fit MSD')
```

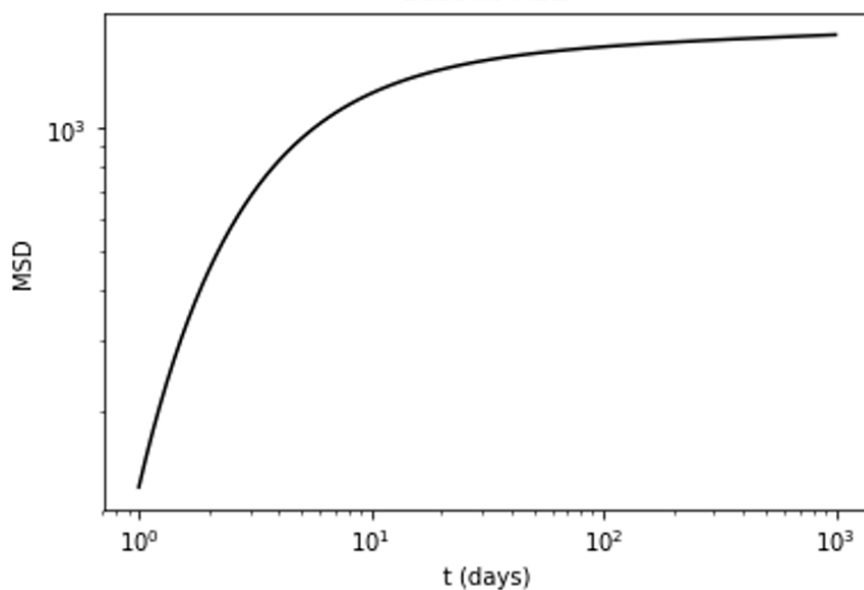


Figure 2.4. MSD plot for Eq. (2.29) with, $a = 3708.31$, $\mu = 1.02$, and $\beta = 2.43$.

2.4.2 Varying MSD Parameters

It is very important to note that the shape of the theoretical MSD is very sensitive to changes in the values of parameters μ and β . Consider, for instance, the effect of the plot for Eq. (2.29) when slightly varying the value of μ equal to $\pm 1 - \sigma$ from the fit, where σ is the standard deviation. Keeping a and β parameters fixed, Fig. 2.5 shows changes in the slope of the upper part of the MSD curve for, $\mu = 1.02 \pm 0.03$, using the following Python code.

Varying μ

```
In: # plot MSD - varying mu

t = np.logspace(0, 3, 100) # equally log-spaced grid in time,
m1 = msd(t, a, beta, mu-err_mu)
m2 = msd(t, a, beta, mu+err_mu)

plt.plot(t, m0, color='k', ls='-')
plt.plot(t, m1, color='k', ls='--')
plt.plot(t, m2, color='k', ls='--')
plt.xscale("log")
plt.yscale("log")
plt.xlabel("t (days)")
plt.ylabel("MSD")
plt.title(r"Varying  $\mu$ : %.2f  $\pm$  %.2f" % (mu, err_mu))

Out: Text(0.5,1,'Varying  $\mu$ : 1.02  $\pm$  0.03')
```

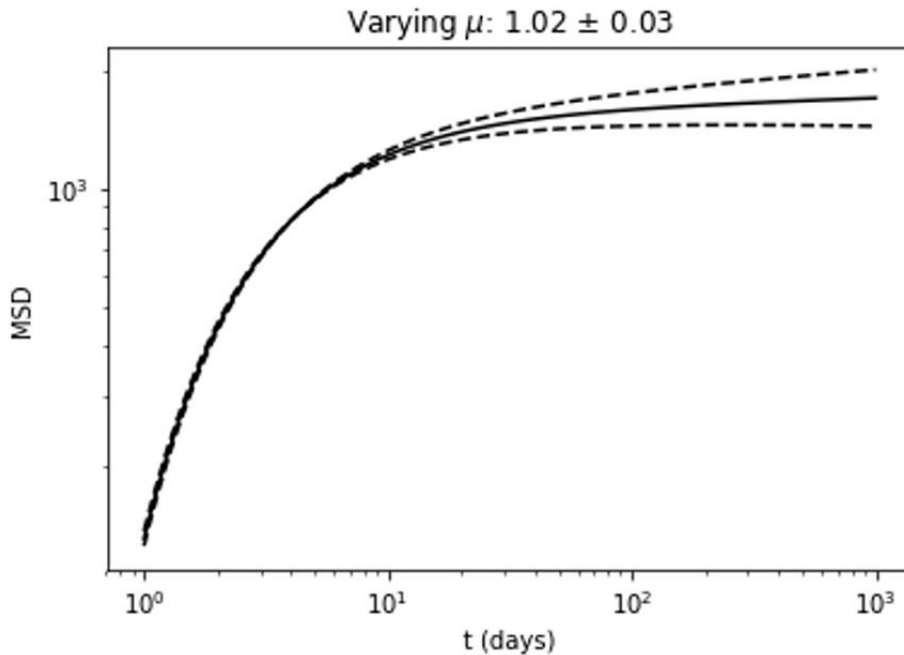


Figure 2.4. Variations of the plot for Eq. (2.29) for parameter values, $\mu = 1.02 \pm 0.03$, with fixed a and β .

On the other hand, with a and μ fixed, minimal changes in the value of β of values, $\pm 1 - \sigma$, i.e., $\beta = 2.43 \pm 0.56$, raises or lowers the curve as shown in Fig. 2.5. This is implemented by the following code.

Varying β

In: # plot MSD - varying beta

```
t = np.logspace(0, 3, 100) # equally log-spaced grid in time, in
m1 = msd(t, a, beta-err_beta, mu)
m2 = msd(t, a, beta+err_beta, mu)

plt.plot(t, m0, color='k', ls='--')
plt.plot(t, m1, color='k', ls='--')
plt.plot(t, m2, color='k', ls='--')
plt.xscale("log")
plt.yscale("log")
plt.xlabel("t (days)")
plt.ylabel("MSD")
plt.title(r"Varying  $\beta$ : %.2f  $\pm$  %.2f" % (beta, err_beta))
```

Out: Text(0.5, 1.0, 'Varying β : 2.43 $\pm$ 0.56')

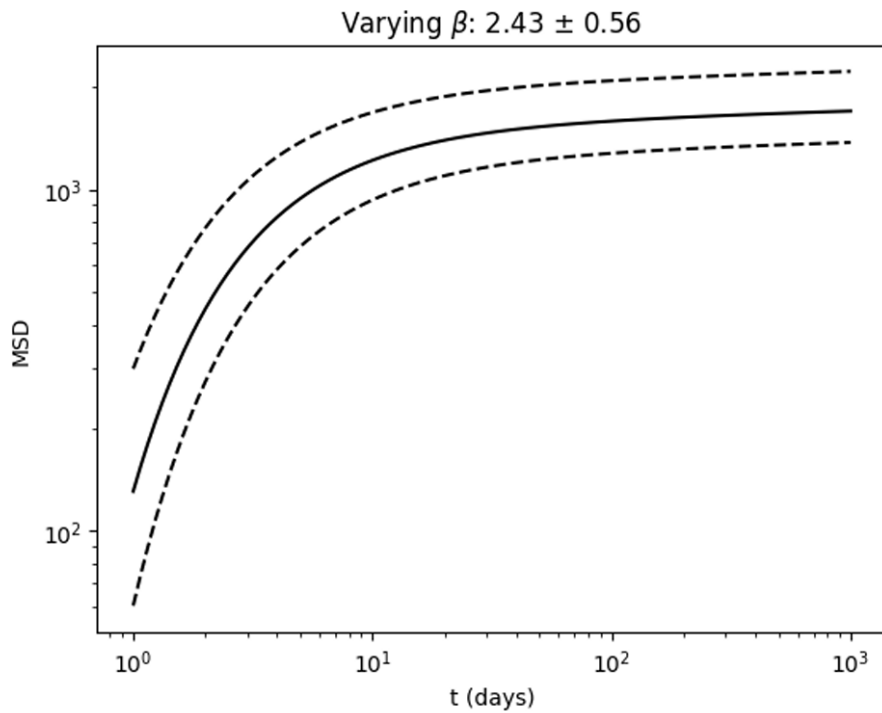


Figure 2.5. Variations of the plot for Eq. (2.27) for parameter values, $\beta = 2.43 \pm 0.56$, with fixed a and μ .

Detrending Datasets

A given dataset is often the outcome of a deterministic part mixed with random fluctuations about the deterministic component. The physics of the deterministic part is often much easier to understand and may even obey known equations. In certain datasets, the impact of the deterministic factor is so dominant that the need for detrending is important to isolate the role of the fluctuating part. Understanding the random fluctuations, sometimes referred to as noise, can oftentimes be challenging. To investigate the memory properties that could be present in a dataset, the isolation of the random fluctuations from the deterministic contribution is often needed. Separating the deterministic part from the random fluctuations is often referred to as detrending. There are many ways of detrending a dataset but, as examples, we will discuss in this chapter four methods: (a) using best-fit curves; (b) employing moving averages; (c) looking at the power spectral density; and (d) applying a cycle averaging analysis which is also called fingerprinting. Note, however, that not all datasets need detrending and a judgment has to be made if it should be applied to certain physical systems depending on the aim of the investigation.

3.1 Best-fit Curves

Using best-fit curves to detrend a dataset is probably the easiest to understand. Given a plot of the dataset, we could superimpose a line or a curve that best captures the dataset. By subtracting from the raw values of the dataset the values of the best-fit curve, we generate fluctuating deviations of the dataset from the best-fit curve. These series of numbers deviating from the best-fit curve could be plotted and investigated for possible memory properties using the stochastic model discussed in Chapter 2.

Consider as an example the daily record of the global atmospheric carbon dioxide (CO_2) concentration measured at the Mauna Loa Observatory in Hawaii since 1958 (www.esrl.noaa.gov). The plot of the CO_2 concentration, known as the Keeling curve in honor of Charles David Keeling (1928-2005), shows oscillations and periodicity as a function of changes in seasons and differences between night and day. The weekly mean CO_2 data from May 1974 to August 2017, is shown as the red graph in Figure 3.1.

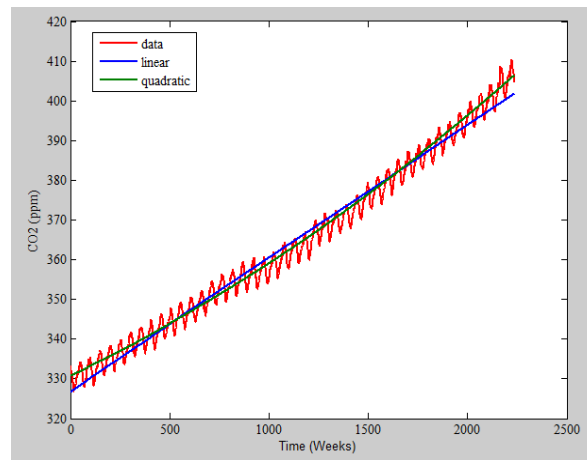


Figure 3.1. Weekly mean of CO_2 concentrations (red) from May 1974 to August 2017 with a linear fit (blue) and a quadratic fit (green).

One way of detrending is to put a linear fit (blue line), or alternatively a quadratic fit (green line), for the data in red. Each CO₂ data point can then be subtracted from the linear fit generating fluctuations as shown in Fig. 3.2 (blue color). For comparison, subtracting the datapoint values from the quadratic fit also gives rise to the fluctuations in Fig. 3.2 (green color). The mean square deviation of the fluctuations in Fig. 3.2 can then be evaluated as discussed in Chapter 4.

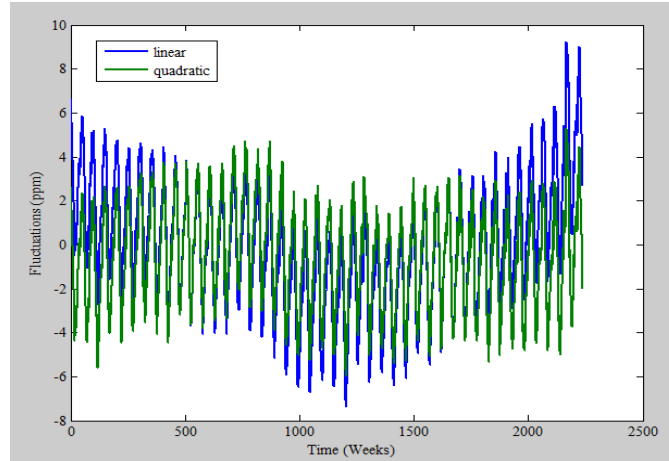


Figure 3.2. Fluctuations obtained by subtracting the CO₂ data points from the linear fit (blue) and quadratic fit (green).

3.2 Moving Averages

Detrending a dataset using moving averages is conceptually straightforward. Consider as an example the time series for the monthly mean for sunspot numbers from solar cycle 1 to solar cycle 24. Figure 3.3 shows the raw data as gray dots and the 7-month average as blue curves.

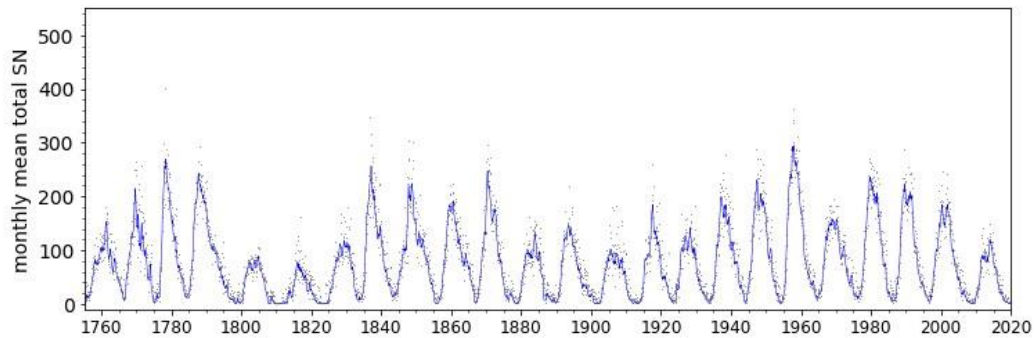


Figure 3.3. Monthly sunspot numbers (gray dots) and 7-month moving average (blue curves) from solar cycle 1 to solar cycle 24. Each solar cycle lasts 11 years on the average.

Subtracting the raw data from the moving average yields the fluctuations which becomes the working data for investigation. The fluctuations or residuals for a moving average of 7 months is shown in Fig. 3.4. The residuals have a mean of 5.87 and standard deviation of 20.75. The mean square deviation for the residuals is then plotted against time and compared with a theoretical MSD.

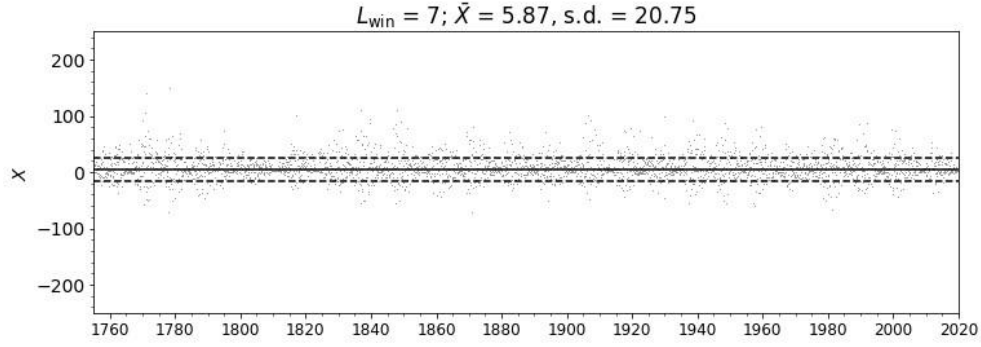


Figure 3.4 The residual time series for monthly sunspot numbers with a mean of 5.87 and standard deviation of 20.75.

3.3 Power Spectral Density Analysis

The deterministic component of a dataset sometimes dominate a signal and the power spectral density (PSD) analysis helps separate the deterministic part from the random fluctuations (Guler et al 2005, Peng et al 2014). For instance, given a time series, one can take its Fourier transform to obtain the frequency domain. Consider the time series for the population of algae in experiment C9 of the paper by Blasius et al (2020) on predator-prey systems. The PSD of the time series can be calculated by taking the normalized squared magnitude of the frequency domain (Fig. 3.5). From a plot of the PSD versus frequency, the peaks occur at dominant frequencies (periodic component) of the signal. To remove the dominant or deterministic part, a PSD threshold can be specified so that we can identify the frequencies of the peaks that go beyond the threshold. In Fig. 3.5, we illustrate four possible thresholds at 1.0, 0.2, 0.3 and 0.5. We then set to zero the frequencies corresponding to the peaks above the chosen threshold. By applying an inverse Fourier transform, we reconvert the filtered signal to the time domain where only the desired fluctuations remain (see, Fig. 3.6, bottom).

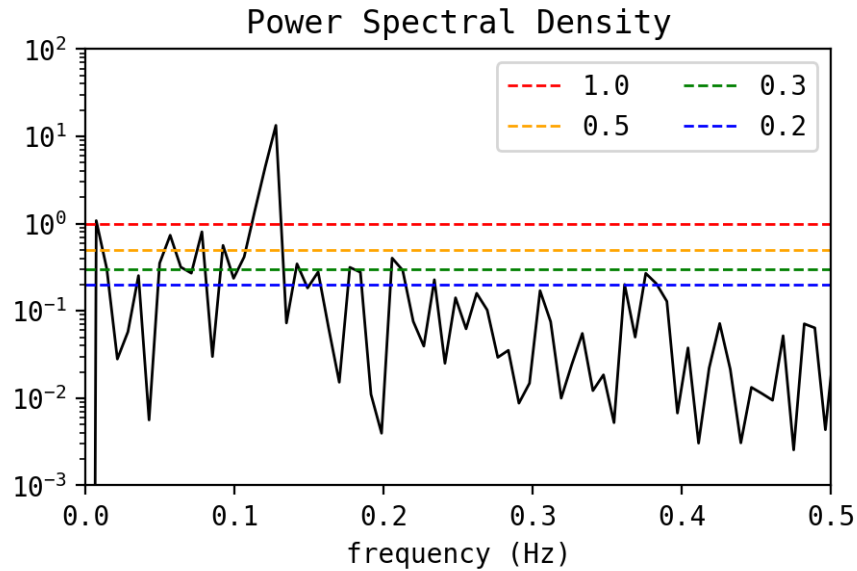


Figure 3.5. Applying a Fourier transform to a time series gives the power spectral density versus frequency plot. The dominant prominent peaks can be removed by choosing a cut-off value. Different possible cut-offs are: 1.0, 0.2, 0.3, or 0.5.

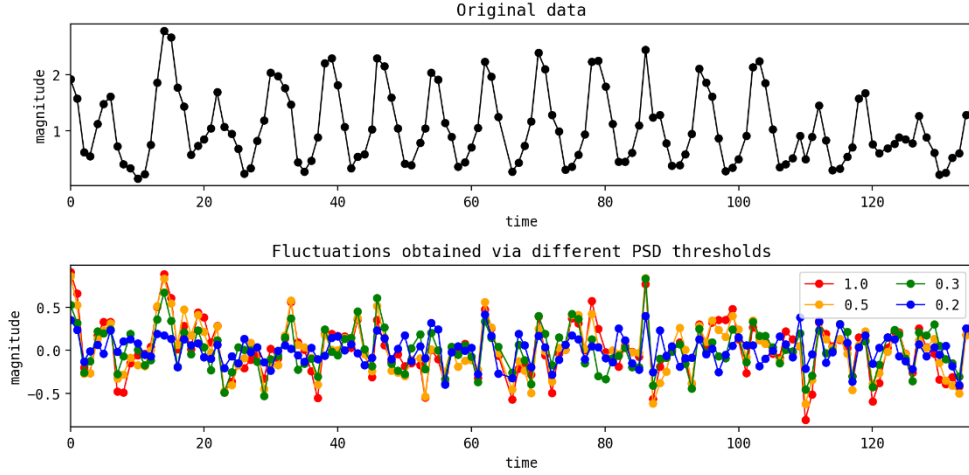


Figure 3.6 The original fluctuating dataset (top). The dominant deterministic part is removed using the power spectral density analysis leaving behind a seemingly random fluctuation as remnants (bottom). The colors of the fluctuations correspond to the different possible cut-offs (1.0, 0.2, 0.3, or 0.5, see, Fig. 3.5) which eliminate the frequency of the dominant or deterministic component.

3.4 Cycle Averaging Analysis: Fingerprinting

Another method of detrending is the cycle averaging analysis which is also referred to as fingerprinting a repeating cycle of a dataset (Jutras and Howard 2015, Cole et al 2009). The average of the repeated cycle is then subtracted from the raw data leaving only the fluctuations about the average. The fluctuations can then be investigated for possible memory inherent in the system.

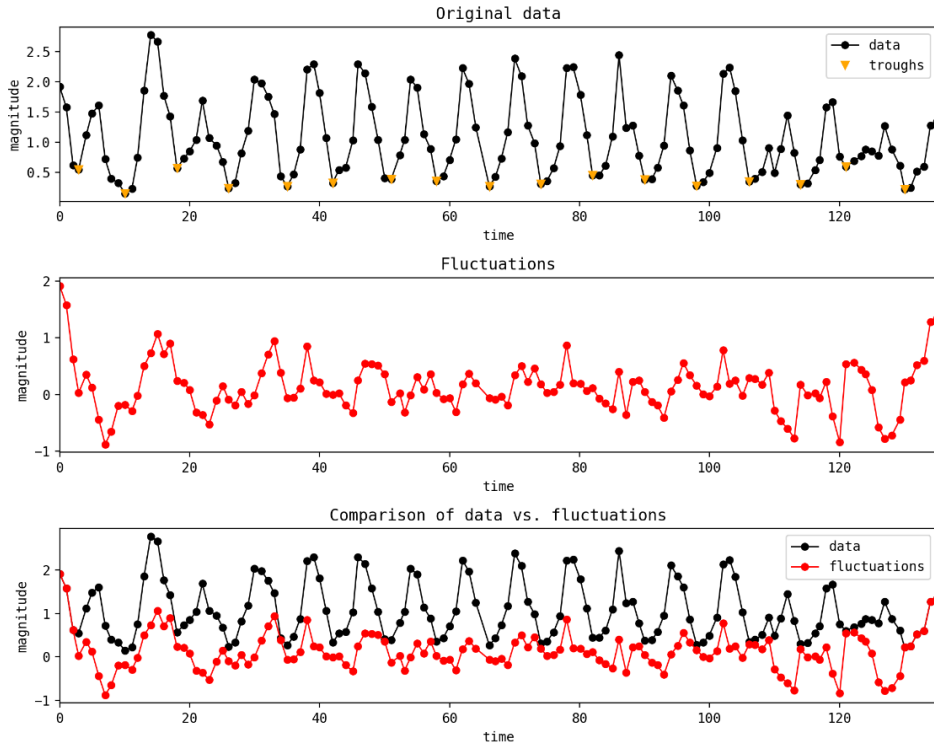


Figure 3.7. The original fluctuating dataset exhibiting a dominant cycle (top). The remaining fluctuation after the dominant deterministic part is removed (middle). Comparison of the original data and the remaining fluctuations (bottom).

Consider a fluctuating cyclic signal shown in Figure 3.7 for the algae population in experiment C9 of a predator-prey system (Blasius et al 2020). We can identify and separate each cycle by cutting them at the trough indicated by the yellow dots in Fig. 3.7 (top). To apply a cycle averaging, we collect all the cycles, or segments, by aligning their maxima. The alignment of maxima may require padding or adding zeroes at the start of some segments. The lengths of all segments are then made equal by also adding or padding zeroes at the end of the cycle (Fig. 3.8, middle). The fingerprint of all the cycles would be the segment mean calculated as the average of all segments equalized in length with aligned maxima (Fig. 3.8, right). The fluctuations of the signal are then obtained by subtracting the mean or fingerprint from each of the original segments (Figure 3.7, middle and bottom). Note, however, that before subtracting the mean, the paddings from the segments are removed, as well as the values of the segment mean at the padding indices, so no new data points are added to the fluctuations. The next step is to get the mean square deviation of the fluctuations as discussed in Chapter 4.

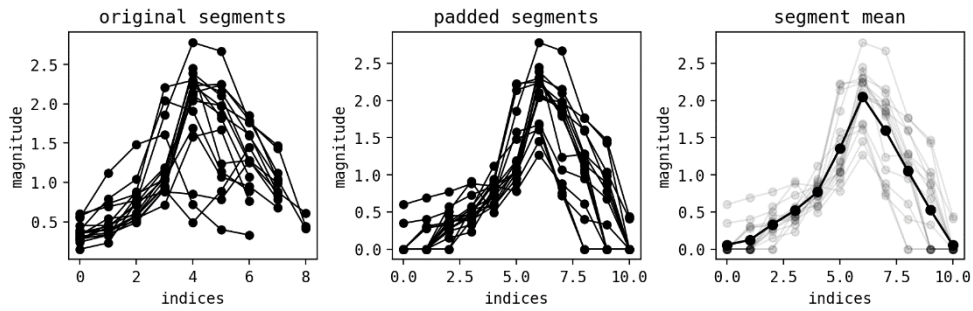


Figure 3.8. The original segments or cycles are collected in preparation for averaging (left). The maxima of the segments are aligned by padding or adding zeroes at the start. The length of all segments is also equalized by padding the end of the segments (middle). The cycle average or fingerprint (right).

Matching Theory with Empirical Data

This chapter bridges the non-Markovian stochastic model discussed in Chapter 2 to actual fluctuations found in various empirical datasets. From the detrended dataset (Chapter 3), we evaluate the mean square deviations of the fluctuations which we shall refer to as the empirical MSD. We then proceed to match the MSD derived from the dataset with a theoretical MSD from the memory modified white noise (MMWN) framework. The MMWN approach has a large collection of different memory behavior to choose from (Table 2.1) to match the empirical MSD. This method often uncovers memory properties which may not be initially obvious from the dataset. Knowing the MSD allows us to obtain the probability density function, Eq. (2.26). A theoretical and empirical matching can also be done between the probability distribution density obtained from the dataset and the theoretical PDF. Further added insights into the behavior and dynamics of the system being investigated can be mined from the PDF as illustrated in Chapter 6.

4.1 Empirical Mean Square Deviation

Consider, for example, a time series with N datapoints, $\{X(t_k)\}$, with $k = 0, 1, 2, \dots, N - 1$. We can introduce a lag time Δ which is defined as the interval between any pair of datapoints and evaluate the deviation (Fig. 4.1), or difference in fluctuation, between these two data points as,

$$DX(\Delta; t_{\text{ref}}) = X(t_{\text{ref}} + \Delta) - X(t_{\text{ref}}) . \quad (4.1)$$

Here, t_{ref} is a reference time at which the difference is measured with $\Delta < N$. These differences in values are important in evaluating the empirical mean square deviation and probability displacement distribution of the dataset.

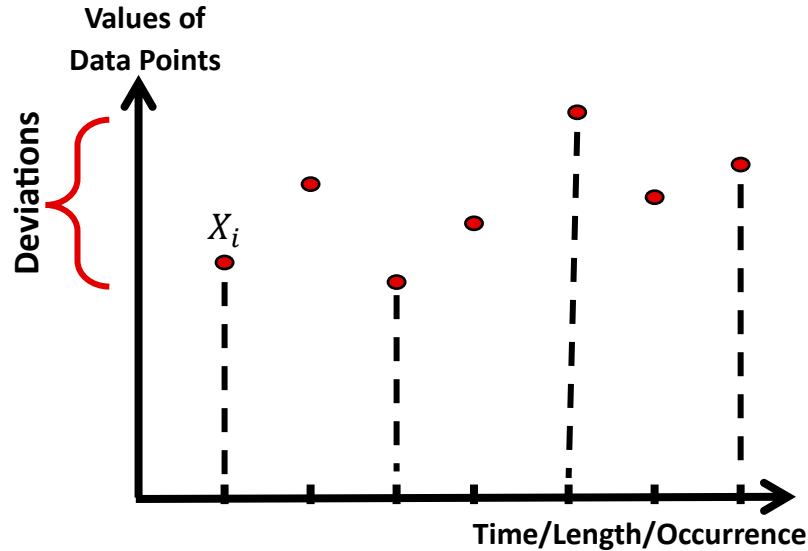


Figure 4.1. The difference in values of any pairs of points separated by an interval Δ are evaluated for all pairs throughout the dataset. For a time series, this interval is called lag time. For other physical systems where measurements may be in length, or sequence of events, the x -axis may be labelled occurrence numbers where the interval Δ between pairs of datapoints may be called occurrence interval.

The empirical mean square deviation for the dataset, denoted by $M_X^{\text{emp}}(\Delta)$, can be evaluated using Eq. (4.1). This is done by squaring the differences and taking an average over all pairs of data points as,

$$M_X^{\text{emp}}(\Delta) = \frac{1}{N - \Delta} \sum_{k=0}^{N-1-\Delta} [DX(\Delta; t_k)]^2 \quad (4.2)$$

Equation (4.2) gives only one MSD value corresponding to lag time Δ . (See Figure 4.2). Increasing the lag time by multiples of Δ generates many values and a plot of $M_X^{\text{emp}}(\Delta)$ versus lag time can be generated. A cut-off for the maximum lag time, however, must be imposed since larger lag time values sample lesser data points resulting in lesser statistical significance. It is advisable to evaluate the MSD only up to $\sim N_{\text{obs}}/2$ so that all data points are essentially sampled.

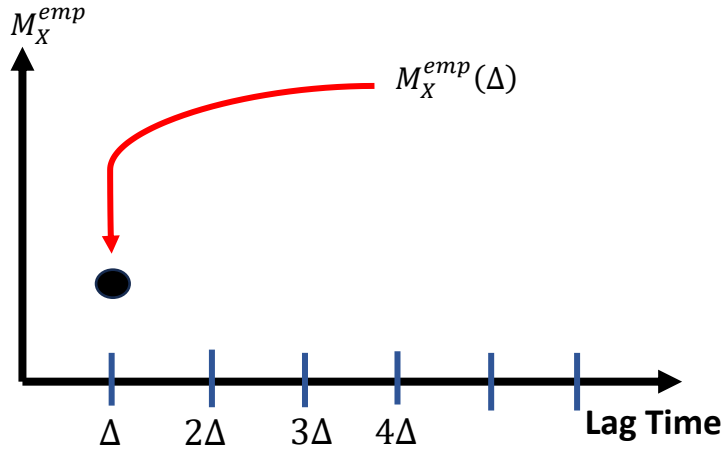


Figure 4.2. The average, Eq. (4.2), gives only one point in the plot for M_X^{emp} corresponding to one Δ . Applying Eq. (4.2) for increasing multiples of Δ generates the empirical MSD curve.

4.2 Ergodic Systems

Many datasets, as exemplified by single particle tracking experiments, are the outcome of a few but very long time series. Since there could be a very limited number of experiments, it is not possible to have an ensemble average of the physical observables involved. Time averaging of the datasets, however, is possible as illustrated by Eq. (4.2). In matching theoretical and empirical MSDs, we are essentially comparing plots generated by an ensemble average Eq. (2.25) using a probability density function with that obtained from Eq. (4.2) using time averaging for the empirical dataset. The equality between the two MSDs assumes an Ergodic system which equates an ensemble average with a time average for a sufficiently long time (van Kampen 2007, Khinchin 2003). We may essentially write, $M_X(t) = M_X^{\text{emp}}(\Delta)$, from Eqs. (2.25) and (4.2), for a sufficiently long time averaged mean square displacement (Metzler et al 2014). In practice, such as in single particle tracking experiments, the time is just the duration of observation for the entire trajectory of the traced particle which allows evaluation of different observables that provide insights into the dynamics of the system. There are, in fact, many examples of ergodic systems which include Brownian motion, fractional Brownian motion, diffusion in fractal environments, and some of those referred to as anomalous diffusion (Metzler et al 2014, S. Thapa et al 2018). In the next section, we view the systems to be essentially ergodic as we discuss the matching of theoretical and empirical MSDs.

4.3 Matching Mean Square Deviations

For a theoretical model to be acceptable, it should agree and be consistent with the recorded experimental measurements of a given phenomenon. Empirically, for a stochastic system, one of the most direct and simplest quantity that can be derived is the mean square deviation of a fluctuating observable as embodied in Eq. (4.2). Hence, for a stochastic phenomenon, one of the fundamental and basic criteria for consistency between theory and experiment is a comparison at the level of mean square deviations which is also referred to as the second order structure function. Generating plots to compare and match theoretical and empirical MSDs can be aided by the SciPy package in Python (Virtanen et al 2020) for 2 parameters, and LMFit for 3 parameters. If unfamiliar with these packages, a trial-and-error approach, i.e., changing the values of the theoretical MSD parameters to improve the match, can always be resorted to as a first approximation. Beyond this, we would like to discuss in this section two items we could apply to the theoretical MSD to improve its fitting with an empirical MSD without violating the mathematical foundations of its derivation. The first item is the role of the function $g(t)$ in Eq. (2.25), and the second is the rescaling of time. Both of these can also be applied simultaneously.

4.3.1 Modulating the Amplitude

The function $g(t)$ modulates the amplitude of the random fluctuation as seen in Eq. (2.3). It appears at the level of the theoretical MSD as (see, Eq. (2.25)),

$$M_X(t) = g(t)^2 \int_0^t [f(t-t') h(t')]^2 dt' . \quad (4.3)$$

In principle, we have the freedom to choose the form of $g(t)$ since it sits outside of the integral. For all entries in Table 2.1, we just let $g(t) = 1$. In many applications, we can set $g(t)^2$ as some constant which effectively raises or lowers the theoretical MSD plot without changing its shape. This is illustrated in Fig. 4.3 where an MSD plot is raised or lowered by multiplying with a constant κ . The parameter κ is sometimes referred to as a normalization constant and the choice, $g(t)^2 = \kappa$, is often helpful.

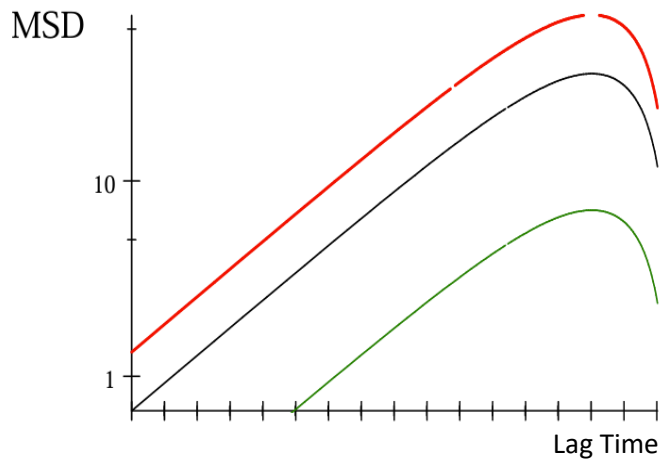


Figure 4.3. The theoretical MSD plot can be raised or lowered without changing its shape by multiplying it by a constant: $\kappa(\text{MSD})$. Black ($\kappa = 1$); Red ($\kappa = 2$); Green ($\kappa = 0.2$).

Let us next consider a case where, $g(t) \neq \text{constant}$. From Eq. (4.3), together with entry number (11) of Table 2.1, the mean square displacement would acquire the form,

$$M_X(t) = g(t)^2 \left[\frac{\sqrt{\pi} \Gamma(\mu) \sin\left(\frac{at}{2}\right) J_{\mu-\frac{1}{2}}\left(\frac{at}{2}\right)}{t^{\frac{1}{2}-\mu} a^{\mu-\frac{1}{2}}} \right]. \quad (4.4)$$

An appropriate choice of $g(t)$ can be useful in matching Eq. (4.4) with an empirical MSD with an odd shape. Let us take as a specific example the form,

$$g(t)^2 = \exp[b \sin(ct)], \quad (4.5)$$

where parameters, b and c , can assume any value. The MSD, Eq. (4.4), becomes,

$$M_X(t) = \exp[b \sin(ct)] \left[\sqrt{\pi} \Gamma(\mu) \left(\frac{t}{a}\right)^{\mu-\frac{1}{2}} \sin\left(\frac{at}{2}\right) J_{\mu-\frac{1}{2}}\left(\frac{at}{2}\right) \right]. \quad (4.6)$$

The plot of Eq. (4.6) is shown in Fig. 4.3 where a comparison with the case, $g(t) = 1$, or $c = 0$ in Eq. (4.5), is shown.

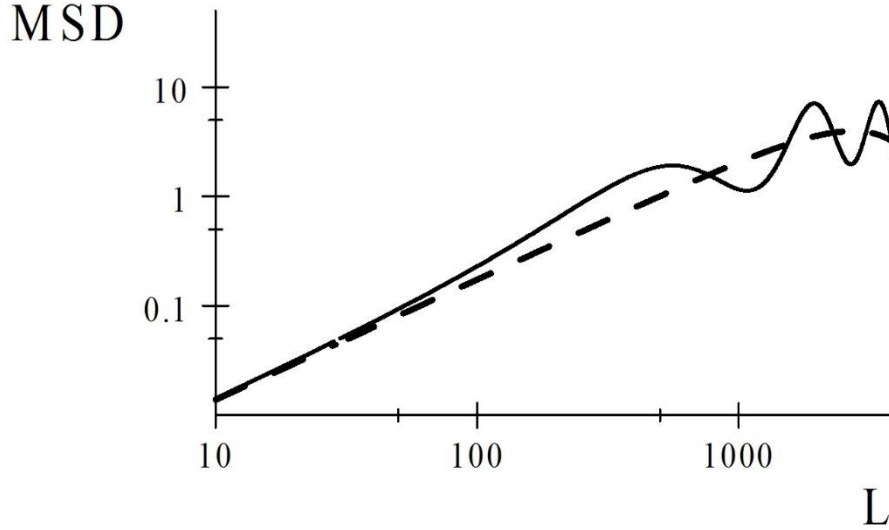


Figure 4.3. Plot of MSD, Eq. (4.6), for, $c = 0$ (dashed line); and $c = 0.0041$ with $b = 0.35$ (solid line). Both plots have parameter values: $\mu = 0.55$ and $a = 0.0008$.

The MSD, Eq. (4.6), was actually used in investigating fluctuating values of the diffusion coefficient for proteins in aqueous solutions. The diffusion coefficients varied as protein length increases with the number of amino acids (Tyn and Gusek 1990, Barredo et al 2018). The role of $g(t)^2$ given by Eq. (4.5) becomes important in attaining a close match of Eq. (4.6) with the empirical MSD as shown in Fig. 4.4. Had the value $g(t)^2 = 1$ been used (dashed line of Fig. 4.3), there was no way to achieve a closer match between the theoretical and empirical MSD (Fig. 4.4).

The advantage of introducing a non-constant time-dependent $g(t)^2 \neq 1$ is that it allows us to capture better the shape of an empirical MSD. We have the freedom to choose the specific form of $g(t)^2 \neq 1$. The downside, however, is that we end up introducing more parameters to determine in our theoretical MSD. In the example

above, Eq. (4.5), we added two more parameters b and c . We then end up with four parameters together with the original μ and a of entry (11) in Table 2.1 as seen in $M_X(t)$, Eq. (4.6).

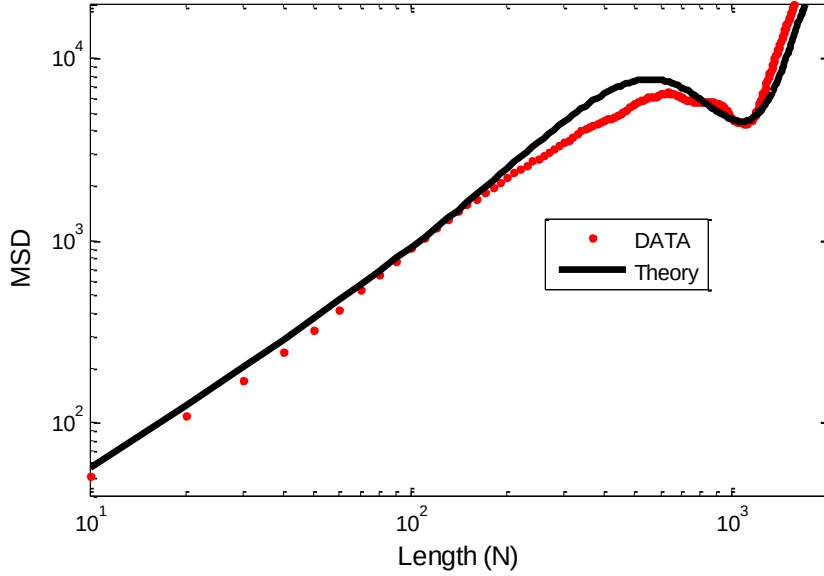


Figure 4.4. Matching the theoretical MSD (black curve) with an empirical MSD (red curve). The theoretical MSD is given by, $3600 M_X(t)$, where $M_X(t)$ is Eq. (4.6) with parameter values, $\mu = 0.55$, $a = 0.0008$, $b = 0.7$, and $c = 0.0041$.

4.3.2 Rescaling Time

The initial step in matching the MSDs is to choose a theoretical MSD, such as those in Table 2.1, which captures the general shape of the empirical MSD. The comparison between theoretical and empirical MSDs is facilitated when the two MSDs overlap. In certain occasions, there is a need to move the theoretical MSD to the left or to the right to achieve a better overlap of MSD plots. Moving the theoretical MSD plot sideways without changing its shape can be done by rescaling the time of $M_X(t)$ that appears in Table 2.1. In particular, we replace the time t by t/t_c where t_c is sometimes referred to as the characteristic time. Depending on the value of t_c , the theoretical MSD plot moves to the right or left without altering its shape (Fig. 4.5).

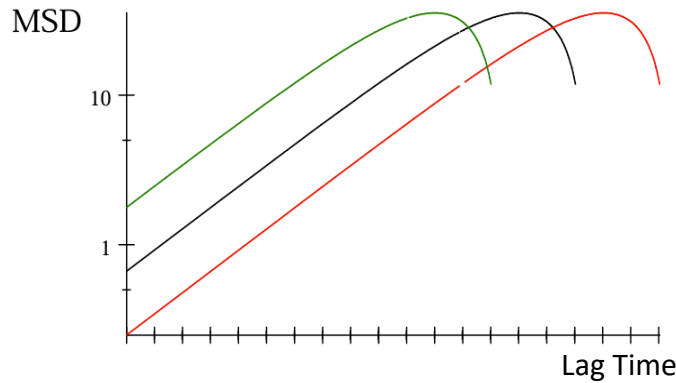


Figure 4.5. Rescaling time as t/t_c moves the theoretical MSD plot right or left without changing its shape. Black ($t_c = 1$); Green ($t_c = 0.5$); Red ($t_c = 2$).

4.4 Empirical Probability Displacement Distribution

For a time series with N datapoints, $\{X(t_k)\}$, with $k = 0, 1, 2, \dots, N - 1$, an empirical probability displacement distribution can be evaluated and plotted. This distribution is evaluated by first choosing a lag time, $\Delta < N$, which is the interval between any pair of datapoints. We can then define the difference $DX(\Delta; t_{\text{ref}})$ between all pairs of points in the dataset separated by the interval Δ as given by Eq. (4.1) where, t_{ref} , is a reference time at which the difference is measured. By counting similar values of $DX(\Delta; t_{\text{ref}})$, we can plot a bar graph showing the frequency of appearance of similar values of $DX(\Delta; t_{\text{ref}})$.

For example, consider the recorded light curves of a high-mass X-ray binary WRAY 15-793 shown in Fig. 4.6. For this dataset, the probability displacement distribution is given by the histogram shown in Fig. 4.7. Oftentimes the frequency distribution exhibits a Gaussian curve. We can repeat this procedure for another value of the lag time Δ and produce another plot showing the frequency distribution for the new interval. The dataset may not be a time series but occurrence of a series of events. In this case, we interpret Δ as an occurrence interval (Violanda et al 2019, Roque et al 2024) instead of a lag time.

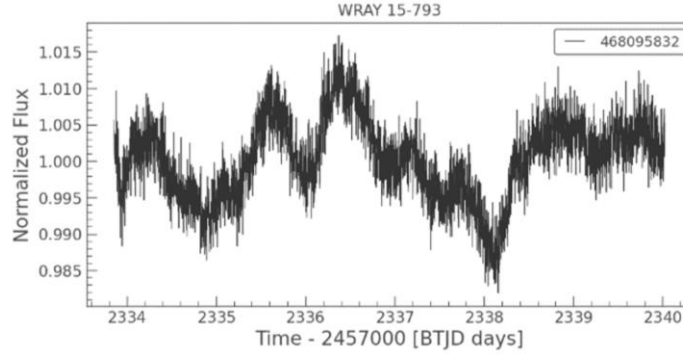


Figure 4.6. Recorded light curves from a high-mass X-ray binary WRAY 15-793.

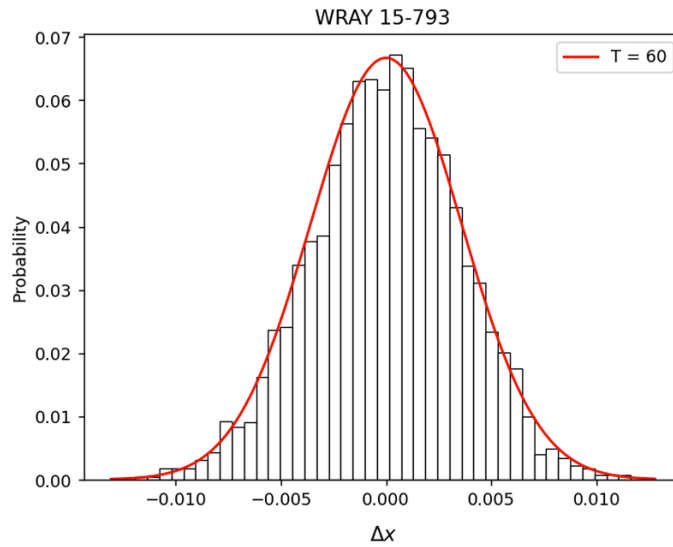


Figure 4.7. Histogram of the frequency of appearance of the same values for $DX(\Delta; t_{\text{ref}})$, Eq. (4.1), obtained from the dataset, Fig. 4.6, for the X-ray binary WRAY 15-793. The red curve is the theoretical PDF, Eq. (2.26), where the mean square deviation $M_X(t)$ is given by entry number 12 of Table 2.1.

4.5 Theoretical and Empirical Probability Distributions

For further validation of the stochastic model used in investigating a dataset, the empirical probability displacement distribution (section 4.4) can also be matched with the theoretical probability density function, Eq. (2.26). The comparison of the probability distributions, however, can reveal information which are not explicitly found in just studying mean square deviations. Firstly, the effect of the drift term does not manifest at the level of mean square deviations (see, e.g., Eq. (2.25)). The drift effects reveal itself noticeably at the level of probability distributions. Secondly, non-Gaussian behavior of the dataset, which is difficult to detect at the MSD level, becomes explicit in probability distributions. We shall touch on these two items in this section.

4.5.1 Drift Term

The effects of a drift term are not evident in the mean square deviation but becomes obvious when we look at probability distributions. Even an initial processing of the dataset, such as getting the probability displacement distribution as discussed in section 4.4, would show that a histogram that is not centered at the origin signals the presence of a drift. A peak of the probability distribution that is either to the left or right of the origin can be handled by the PDF given by Eq. (2.26),

$$P(X_f, X_i; t) = \frac{1}{\sqrt{2\pi M_X(t)}} \exp \left[-\frac{\left(X_i - X_f + \int_0^t a_1(t') dt' \right)^2}{2 M_X(t)} \right], \quad (4.7)$$

where, $\int_0^t a_1(t') dt'$, is the drift term, and $M_X(t)$ contains the memory function. In principle, we could choose any form of $a_1(t')$ suitable to the physical system being investigated. The simplest form for the drift is a constant, i.e., $a_1(t') = k$, which yields, $\int_0^t a_1(t') dt' = kt$. To illustrate its effect, we could plot the PDF for cases where, $kt = 0$, and compare it with a case where, for example, $kt = 4.5$ (see, Fig. 4.8).

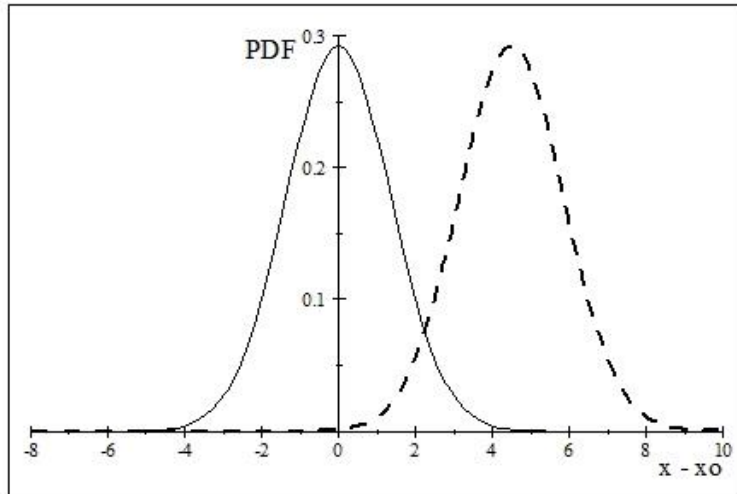


Figure 4.8. PDF, Eq. (4.7), with zero and nonzero drift: $\int_0^t a_1(t') dt' = 0$ (solid line); $\int_0^t a_1(t') dt' = 4.5$ (dashed line). The presence of a drift moves the peak of the PDF away from the origin.

The plot of the PDF in Fig. 4.8 used Eq. (4.7) with an $M_X(t)$ given by entry number 12 of Table 2.1 for, $\mu = 4.64$ and $a = 0.99$, with time set to, $t = 3$. A negative drift term would result in a PDF plot with a peak at the left of the origin.

4.5.2 Non-Gaussian Behavior

Another characteristic of a dataset which is not quite obvious at the level of the MSD is the non-Gaussian property of a fluctuating variable. This is best illustrated by an example where the mean square deviation of the measurements is generally captured by the theoretical MSD plot using Eq. (2.25), but not by the Gaussian PDF, Eq. (2.26), because of the inherent non-Gaussian features of the dataset.

As an example, let us consider as our dataset the recorded time residuals of signals emitted by pulsar B1937+21 for the 820 MHz band from 2008 to 2016 as shown in Fig. 4.9.

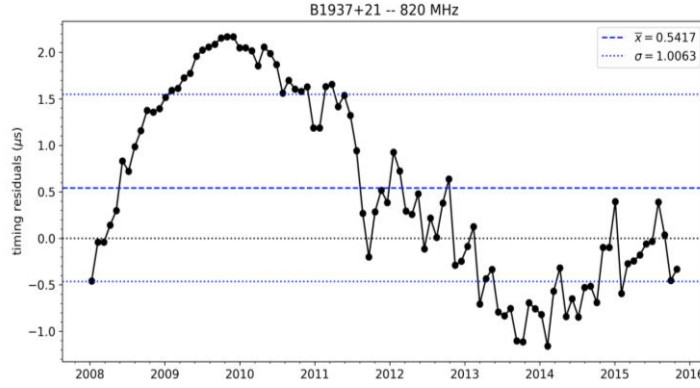


Figure 4.9. Recorded time residuals of the signal emitted by pulsar B1937+21. Dashed blue line is the mean and the horizontal blue dots show the standard deviation.

The empirical mean square deviation M_X^{emp} for the dataset in Fig. 4.9 can then be evaluated using Eq. (4.2) as discussed in section 4.1. The plot of the empirical MSD can then be matched with a theoretical mean square deviation $M_X(t)$ given by entry number 12 of Table 2.1. The result of the MSD comparison is shown in Fig. 4.10.

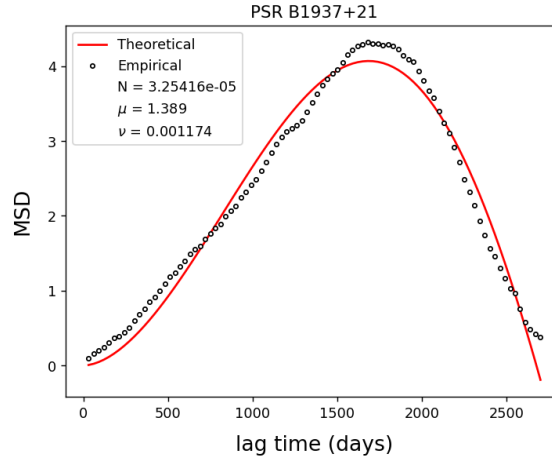


Figure 4.10. The theoretical MSD (red) essentially captures the shape of the empirical MSD (open circles) of the timing residuals (Fig. 4.9) for pulsar B1937+21.

At the MSD level, as shown by Fig. 4.10, we appear to successfully model the stochastic properties of the timing residuals of pulsar B1937+21. There are things, however, not revealed or missed out by the MSD. We also need

to match the empirical probability distribution with the theoretical PDF, Eq. (2.26), using as $M_X(t)$ entry number 12 of Table 2.1 which was used in Fig. 4.10 for the MSD matching. By generating the probability displacement distribution for the dataset as discussed in section 4.4, we quickly find out that the theoretical PDF, Eq. (2.26), fails to capture the shape of the empirical probability distribution. The profile of the empirical probability distribution for the dataset, Fig. 4.9, is shown in Fig. 4.11 for different lag times Δt . Clearly, looking at Fig. 4.11, the theoretical PDF to be used has to be non-Gaussian. The PDF that we have, Eq. (2.26), however, is Gaussian. There is a need, therefore, to extend the theoretical framework to include non-Gaussian cases which is discussed in Chapter 7.

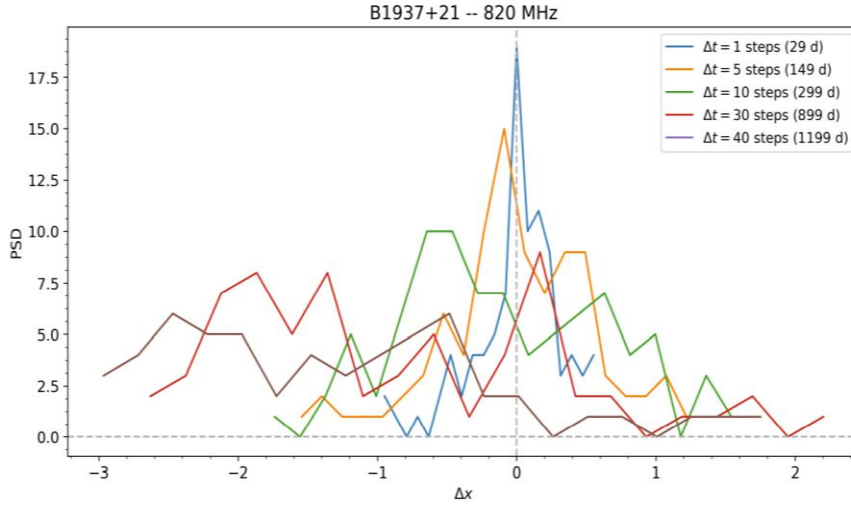


Figure 4.11. The probability displacement distributions for the timing residuals shown in Fig. 4.9 for different lag times Δt .

So why was the MSD matching apparently successful as shown in Fig. 4.10, but not at the level of the probability density function? The reason for this is clarified in Chapter 7 where the MSD for non-Gaussian processes differs only by a constant factor from the MSD of the Gaussian case.

First Passage Time Density

In the previous chapters, we discussed a framework for analyzing big datasets by obtaining a probability density function to describe a fluctuating observable. The question often asked is when this stochastic variable would hit a threshold or a critical value for the first time. To determine when a variable first hits or passes a certain value is called the first passage time (FPT) problem. Knowing the first passage time is crucial in many investigations ranging from heterogeneous and disordered media (Vaccario et al 2015, Kalantar and Segal 2018), the treatment of cancer (Hufton et al 2020), neuronal models of the brain (G. D'Onofrio et al 2018), all the way to epidemiology (Castro et al 2018) and financial markets (Lipton and Kaushansky 2020), among many others. Many studies on the FPT, however, make use of Markovian (memoryless) processes (Lipton and Kaushansky 2020, Hartich and Godec 2019) which are unable to handle systems with highly correlated values of the observables. Real-world models need to incorporate memory effects as an offshoot of a system's strong interaction with the environment. An exact analytical result for the FPT, however, is rarely available such that numerical methods and approximations are often employed to approach nontrivial systems (Hufton et al 2020, Castro et al 2018, Guérin et al 2016). In this chapter, we obtain an exact analytical expression of the first passage time density for a class of memory modulated white noise processes discussed in the previous chapters. We begin by first obtaining the probability density function that satisfies a Dirichlet boundary condition. The corresponding first passage time density is then obtained exactly, but general enough to accommodate stochastic processes with a variety of memory behavior. We give as examples, the power law diffusion, fractional Brownian motion, and the Great Barrier Reef.

5.1 Dirichlet Boundary Conditions

Many natural systems require boundary conditions such as those in domains with absorbing boundaries. We extend the applicability of the large group of white noise processes with memory by considering a random variable $X(\tau)$ which evolves from an initial point, $X(0) = X_i$, at time $\tau = 0$, and treat the threshold value as a boundary, or “cliff point” at, X_c at time t . Pictorially, we can represent the threshold X_c as a boundary where the PDF vanishes, i.e., $p(X = X_c, X_i; t) = 0$. To obtain the PDF satisfying this boundary condition, we use the method of images (Bernido and Carpio-Bernido 2002, 2015) where we assume an image point at a distance $2X_c - X_i$ from the origin (Fig. 5.1).

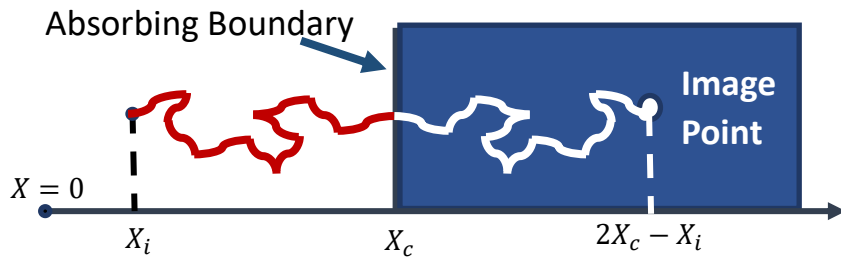


Figure 5.1. Two paths are considered: one from X_i and the other from an image point at $2X_c - X_i$.

Let us next consider two fluctuating paths. One path, X^a , originates from X_i and the other path X^b coming from an image point, $2X_c - X_i$. For both stochastic paths to end at the same final endpoint, X_f , we pin down the final point using the Dirac delta function (see, e.g., Figs. 2.1 and 2.2), and write the linear combination,

$$L = \delta(X^a - X_f) - \delta(X^b - X_f) . \quad (5.1)$$

The path X^a is given by (see, e.g., Eq. (2.1)),

$$X^a = X_i + \int_0^t a_1(t') dt' + \int_0^t a_2(t, t') \omega(t') dt' , \quad (5.2)$$

which describes white noise fluctuations modulated by $a_2(t, t')$ originating from X_i with a drift term $a_1(t')$. On the other hand, the path X^b is,

$$X^b = 2X_c - X_i - \int_0^t a_1(t') dt' + \int_0^t a_2(t, t') \omega(t') dt' , \quad (5.3)$$

which describes the source of fluctuations from an image point, $2X_c - X_i$, with a drift term of opposite sign. The probability density function is obtained by summing-over-all-paths emanating from these two source points that end at X_f . This is done by integrating Eq. (5.1) using the Gaussian white noise measure $d\mu(\omega)$. Using Eqs. (5.1) to (5.3), we have,

$$\begin{aligned} p(X_f, X_i; t) &= \int L d\mu(\omega) \\ &= \int \left[\delta \left(X_i + \int_0^t a_1(t') dt' + \int_0^t a_2(t, t') \omega(t') dt' - X_f \right) \right. \\ &\quad \left. - \delta \left(2X_c - X_i - \int_0^t a_1(t') dt' + \int_0^t a_2(t, t') \omega(t') dt' - X_f \right) \right] d\mu(\omega) \end{aligned} \quad (5.4)$$

Now, using the Fourier representation of the delta function, we can rewrite Eq. (5.4) as,

$$\begin{aligned} p(X_f, X_i; t) &= \frac{1}{2\pi} \int \left\{ \int_{-\infty}^{+\infty} \exp \left[ik \left(X_i - X_f + \int_0^t a_1(t') dt' \right. \right. \right. \\ &\quad \left. \left. + \int_0^t a_2(t, t') \omega(t') dt' \right) \right] dk \\ &\quad - \int_{-\infty}^{+\infty} \exp \left[ik \left(2X_c - X_i - X_f - \int_0^t a_1(t') dt' \right. \right. \\ &\quad \left. \left. + \int_0^t a_2(t, t') \omega(t') dt' \right) \right] dk \right\} d\mu(\omega) \end{aligned} \quad (5.5)$$

We can factor out the terms dependent on the white noise variable $\omega(\tau)$ to facilitate integration over the Gaussian white noise measure $d\mu(\omega)$, i.e.,

$$\begin{aligned}
p(X_f, X_i; t) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} dk \left[e^{ik(X_i - X_f + \int_0^t a_1(t') dt')} - e^{ik(2X_c - X_i - X_f - \int_0^t a_1(t') dt')} \right] \\
&\quad \times \int \exp \left[ik \int_0^t a_2(t, t') \omega(t') dt' \right] d\mu(\omega). \tag{5.6}
\end{aligned}$$

If we let $\xi(t') = ka_2(t, t')$, the characteristic functional, Eq. (1.11), allows us to integrate over $d\mu(\omega)$ to get,

Non-Gaussian Stochastic Model

The non-Gaussian stochastic processes presented in this chapter makes use of the Mittag-Leffler measure (Grothaus et al 2014) in infinite dimensional analysis. Parametrizing the fluctuating variable in terms of the generalized Grey noise, we obtain a closed form of the probability density function for a huge collection of varying memory behavior. The corresponding mean square deviation for the process is likewise obtained. The Gaussian white noise stochastic processes discussed in chapter 3 is a special case of this formalism. We begin by introducing the Mittag-Leffler function and the M -Wright function which are essential in constructing the non-Gaussian stochastic framework.

7.1 Mittag-Leffler Function and M -Wright Function

The Mittag-Leffler function $E_\beta(z)$ introduced in 1903 by Mittag-Leffler (1846-1927) has been referred to as the queen function of fractional calculus (Mittag-Leffler 1903). The $E_\beta(z)$ is defined as,

$$E_\beta(z) = \sum_{n=0}^{\infty} \frac{z^n}{\Gamma(\beta n + 1)} \quad (7.1)$$

where, $z \in \mathbb{C}$, $n \in \mathbb{N}$, and $0 < \beta \leq 1$. Equation (7.1) gives us a familiar function when $\beta = 1$, noting that, $\Gamma(n + 1) = n!$. We have,

$$E_1(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!} = e^z \quad (7.2)$$

The case $\beta = 1$, is important in our discussion since this corresponds to a Gaussian stochastic processes using the white noise random variable $\omega(t)$. Closely related to the Mittag-Leffler function is the M -Wright function $M_\beta(z)$. In particular, the Laplace transform of $M_\beta(z)$ gives $E_\beta(z)$, i.e.,

$$E_\beta(-s) = \int_0^{\infty} e^{-s\tau} M_\beta(\tau) d\tau . \quad (7.3)$$

The M -Wright function $M_\beta(z)$ can be written as,

$$M_\beta(z) = \sum_{n=0}^{\infty} \frac{(-z)^n}{n! \Gamma(-\beta n + 1 - \beta)} \quad (7.4)$$

We can also introduce the M -Wright function in two variables $M_\beta(x, t)$. This is given by (Gorenflo and Mainardi 2012),

$$M_\beta(x, t) = t^{-\beta} M_\beta(xt^{-\beta}) . \quad (7.5)$$

Its integral representation is given by,

$$M_{\beta/2}(x, t) = 2 \int_0^{\tau} \frac{e^{-x^2/4\tau}}{\sqrt{4\pi\tau}} t^{-\beta} M_{\beta}(t^{-\beta}\tau) d\tau \quad (7.6)$$

For the case $\beta = 1$, we have the relation (Gorenflo and Mainardi 2012),

$$\frac{1}{2} M_{1/2}(x, t) = \frac{1}{\sqrt{4\pi t}} e^{-x^2/4t}, \quad (7.7)$$

which is the Gaussian density evolving with time with variance $2t$.

7.2 Mittag-Leffler Measure

To discuss non-Gaussian fluctuations, we may use the generalized grey noise ω . To handle and integrate functionals of ω , we would need to define the Mittag-Leffler measure. Mathematically, we define the Mittag-Leffler measure by considering a real separable Hilbert space $\mathcal{H}$. We then define a space which is continuously and densely embedded in $\mathcal{H}$, i.e., $\mathcal{N} \subset \mathcal{H}$. The $\mathcal{N}$ is called a nuclear space and we designate its dual space as $\mathcal{N}'$, such that we have the inclusions, $\mathcal{N} \subset \mathcal{H} \subset \mathcal{N}'$. The Mittag-Leffler measure is then a unique probability measure μ_{β} on $\mathcal{N}'$ and the $d\mu_{\beta}(\omega)$ is defined by the characteristic functional (Grothaus et al 2014),

$$\int e^{i\langle \omega, \varphi \rangle} d\mu_{\beta}(\omega) = E_{\beta} \left(-\frac{1}{2} \langle \varphi, \varphi \rangle \right), \quad (7.8)$$

where $\varphi \in \mathcal{N}$, and $E_{\beta}(z)$ is the Mittag-Leffler function with $0 < \beta \leq 1$. The $\langle \cdot, \cdot \rangle$ denotes the dual pairing between $\mathcal{N}'$ and $\mathcal{N}$. As an example, Eq. (7.8) can be expressed in a more familiar notation for practical physical applications as,

$$\int e^{i \int_0^t \omega(\tau) \varphi(\tau) d\tau} d\mu_{\beta}(\omega) = E_{\beta} \left(-\frac{1}{2} \int_0^t \varphi^2 d\tau \right). \quad (7.9)$$

The characteristic functional Eq. (7.9) is essential in evaluating and obtaining a closed form for the non-Gaussian probability density function that accommodates many possible memory behavior.

7.3 Non-Gaussian Probability Density Function

Using the generalized grey noise $\omega(\tau)$, let us parametrize a fluctuating variable, $X(t) = X_i + \text{fluctuations}$, for a non-Gaussian process as,

$$X(t) = X_i + g(t) \int_0^t f(t - \tau) h(\tau) \omega(\tau) d\tau. \quad (7.10)$$

The exact form of the functions $g(t)$, $f(t - \tau)$, and $h(\tau)$ modulating the random noise ω will depend on the physical system being investigated. The function $g(t)$ controls the amplitude of the fluctuation. The function $f(t - \tau)$ characterizes an elapsed time $(t - \tau)$ and is called a memory function, while $h(\tau)$ is a window function which also modifies the random noise $\omega(\tau)$. In Eq. (7.10), the starting point of the fluctuation is fixed at $X(0) = X_i$, at initial time, $\tau = 0$. The fluctuation, however, could end at any value at final time $\tau = t$. The probability density function (PDF) that the fluctuations acquire a value, $X(t) = X_f$, at some final time $\tau = t$, if it started at X_i , can be evaluated. First, we consider only all those paths which end at X_f . This is described by fluctuations pinned down at X_f at final time $\tau = t$, using the Dirac delta function (see, Fig. 1),

$$\delta(X(t) - X_f) = \delta\left(X_i + g(t) \int_0^t f(t - \tau)h(\tau) \boldsymbol{\omega}(\tau) d\tau - X_f\right). \quad (7.11)$$

Here, we have used Eq. (7.10) to express $X(t)$. The delta function is always zero for all paths described by Eq. (7.10), except those which end at the chosen endpoint X_f . When can then sum, or integrate, over all possible paths satisfying Eq. (7.11) to obtain the probability density function (PDF), $P(X_f, X_i; t)$, given by,

$$P(X_f, X_i; t) = \int \delta(X(t) - X_f) d\mu_\beta(\boldsymbol{\omega}) \quad (7.12)$$

We use the Mittag-Metzler measure $d\mu_\beta(\boldsymbol{\omega})$ since $X(t)$ in Eq. (7.12) contains the generalized grey noise $\boldsymbol{\omega}$ as shown in Eq. (7.10). To evaluate the PDF, we express the delta function in terms of its Fourier representation,

$$P(X_f, X_i; t) = \frac{1}{2\pi} \int \int_{-\infty}^{+\infty} \exp[ik(X(t) - X_f)] dk d\mu_\beta(\boldsymbol{\omega}) \quad (7.13)$$

To facilitate the integration, we rearrange Eq. (7.12) as,

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